

Zeroth-Order Optimization

MSc in Advanced Computer Science

Trinity Term 2026

Candidate number: TODO

Word count: TODO

Contents

Chapter 1

Related Work

1.1 Zeroth-Order Optimization and Evolutionary Strategies

The first evolutionary strategy originated in Rechenberg's experimental work on aerodynamic design. In a wind-tunnel experiment, he repeatedly applied small random mutations to the five joint angles of a segmented plane, retaining a mutation whenever it reduced measured drag. After only around 300 mutations, this process was able to find a minimum-drag configuration ?. Later, he developed the theoretical basis behind the $(1 + 1)$ -ES and the celebrated $1/5$ success rule ?. His was the first of many algorithms which are now known as zeroth-order optimization algorithms, who can query an oracle $f(x)$ but do not have access to its gradient $\nabla f(x)$.

Evolutionary strategies have been developed further since then. Instead of just using one mutation per time step, one could use a larger population size. The question then becomes, how to use the recombine the function evaluations most effectively. This led to ? developing Covariance Matrix Adaptation Evolution Strategy (CMA-ES) that adopts a Gaussian distribution over candidate solutions, gradually moving the mean towards a

weighted average of the best candidates and updating the covariance to align with directions that repeatedly produce improvement.

Next came Natural Evolutionary Strategies (NES) which reduces expected fitness using the score-function identity $\nabla_{\theta} \mathbb{E}_{z \sim p_{\theta}} [f(z)] = \mathbb{E}_{z \sim p_{\theta}} [f(z) \nabla_{\theta} \log p_{\theta}(z)]$? then rescaling this gradient estimate using an approximate Fisher information preconditioner. The main idea is that the important geometry is induced by the KL divergence between two policies, not the raw gradient in θ . While these methods demonstrated remarkable empirical performance, the problem is that this Fisher preconditioner and covariance matrix used in CMA-ES are $d \times d$, which is impractical for large ML models.

1.2 Scalable Evolutionary Strategies

This gave rise to scalable ES. ? directly optimize the smoothed surrogate $f_{\sigma}(x) = \mathbb{E}_{p \sim \mathcal{N}(0, I)} [f(x + \sigma p)]$ of f whose derivative satisfies $\nabla_x f_{\sigma}(x) = \frac{1}{\sigma} \mathbb{E}_{p \sim \mathcal{N}(0, I)} [f(x + \sigma p) p]$. They then use the Monte-Carlo estimator with r parallel workers

$$\frac{1}{r} \sum_{i=1}^r \frac{f(x + \sigma p_i) - f(x - \sigma p_i)}{2\sigma} p_i$$

to repeatedly update the parameters thereby decreasing $f_{\sigma}(x)$. To scale this approach, instead of sharing full perturbations across workers, they use pseudo-random generators to sample perturbations, and then they only need to share the seeds between CPUs, allowing each GPU to reconstruct every perturbation locally. Marking the first time evolutionary strategies was scaled, they were able to use over 1000 parallel workers.

For large ML models with trillions of parameters, 1000 parallel workers is not nearly enough to optimize losses in any reasonable time. The central bottleneck is that, with r parallel workers, each needs to hold $O(r \times d)$ parameters where d is the dimension the each Gaussian perturbations. This is simply impossible to scale to large population

size as d is too large. ? have the solution: instead of using full Gaussian perturbations $P \sim \mathcal{N}(0, I_{d \times m})$, we can use low-rank perturbations $E = \frac{1}{\sqrt{k}}AB^\top$ where $A \sim \mathcal{N}(0, I_{d \times k})$ and $B \sim \mathcal{N}(0, I_{m \times k})$. Then, we can exploit $u(M + \sigma E)^\top = uM^\top + \frac{\sigma}{\sqrt{k}}(uB)A^\top$. They were able to scale evolutionary strategies for the first time to millions of parallel workers, demonstrating that zeroth-order methods really can be used in practice for modern ML models. The central question of Chapter ?? is whether or not this low-rank approach converges.

1.3 Accelerated Optimization

For convex functions, Gradient descent already achieves an $O(L \|x_0 - x^*\|^2 / \varepsilon)$ convergence to an ε -optimal point $f(x) - f(x^*) \leq \varepsilon$, where $x^* \in \operatorname{argmin}_{x \in \mathbb{R}^d} f(x)$. For L -smooth and μ -strongly convex functions, gradient descent achieves $O\left(\frac{L}{\mu} \log\left(\frac{1}{\varepsilon}\right)\right)$ convergence to an ε -optimal point. But can we do better?

? introduced the Heavy Ball Method which achieves the accelerated rate of $O\left(\sqrt{\frac{L}{\mu}} \log\left(\frac{1}{\varepsilon}\right)\right)$ for strongly convex quadratics but may not even converge for general convex f . The algorithm included the momentum term $(x_k - x_{k-1})$,

$$x_{k+1} = x_k - \eta \nabla f(x_k) + \beta(x_k - x_{k-1})$$

Then in his seminal work, ? introduced Accelerated Gradient Descent by forming the “lookahead” term y_k and time-varying $\beta_k = (k-1)/(k+2)$,

$$y_k = x_k + \beta_k(x_k - x_{k-1}), \quad x_{k+1} = y_k - \eta \nabla f(y_k)$$

For general convex f , it converges to an ε -optimal point in $O(1/\sqrt{\varepsilon})$ iterations. By restarting this AGD every $O(\sqrt{L/\mu})$ iterations, Nesterov’s AGD attains the same $O\left(\sqrt{\frac{L}{\mu}} \log\left(\frac{f(x_0) - f^*}{\varepsilon}\right)\right)$ convergence for all μ -strongly convex functions, not just quadratics.

While Nesterov’s AGD is powerful, it is relatively unintuitive, and little is known why β_k is chosen to be so outside of an algebraic trick to telescope a series. [1] introduce linear coupling, which provides a relatively intuitive explanation of acceleration. Instead of a mysterious lookahead point, one maintains a gradient sequence y_k and a mirror sequence z_k . Each are updated at the gradient $\nabla f(x_k)$, but they give differing answers of where to go next. Linear coupling says to go to the convex combination $x_{k+1} = \lambda z_k + (1 - \lambda)y_k$. By carefully tuning the mirror descent and coupling parameters, they show a restarted version of this scheme achieves the same acceleration for convex functions as Nesterov’s AGD. This framework will become the basis for Chapter [2] and the motivation for investigating Zeroth-Order Mirror descent in Chapter [3].

[4] developed an accelerated random gradient-free scheme for smooth convex functions using Gaussian search directions obtaining an expected $O(d^2/T^2)$ rate in their normalization. Typically, they posit, zeroth-order methods take d times longer than their first-order counterparts. They emphasize that accelerated methods are more sensitive to oracle errors because inexact oracle errors can propagate through momentum ([5]). Our analysis in Chapter [2] precisely quantifies this accumulated error in the linear mixing framework.

1.4 Continuous-Time Interpretations of Acceleration

One possible interpretation of acceleration comes from investigating what happens in continuous time. [6] show that the ODE arising from taking the step size in Nesterov’s accelerated gradient descent yields

$$\ddot{X}_t + \frac{3}{t}\dot{X}_t + \nabla f(X_t) = 0$$

It is surprising that a first-order method yields a second-order ODE. They prove that this ODE also achieves $O(1/t^2)$ convergence to an ε -optimal point for convex f . The

constant $3 = 2 + 1$ arising from $(k - 1)/(k + 2) \approx 1 - 3/k$ turns out to be of critical importance. Namely, it is the smallest constant yielding the $O(1/t^2)$ acceleration. Under suitable normalization, they show this ODE approaches Nesterov's AGD in a maximal sense. They show it also converges at rate $O(1/T^3)$ for strongly convex f , and posit a better Lyapunov could yield the same rate for the discrete version, but this remains open.

Future work in this direction includes ? who analyze the Tikhonov regularized version of the previous ODE

$$\ddot{X}_t + \frac{\alpha}{t} \dot{X}_t + \nabla f(X_t) + \frac{c}{t^2} X_t = 0.$$

This second-order ODE has remarkable properties. For $\alpha > 3$, $f(X_t) - f^* = O(1/t^2)$, the trajectory $\|\dot{X}_t\| = O(1/t)$ is bounded, and some subsequence converges to the minimum norm solution $\liminf_{t \rightarrow \infty} \|X_t - x^*\| = 0$. The simple addition of the term $c/t^2 X_t$ to the dynamics provides these extra properties. While we find this result striking and believe discretizing continuous-time dynamics with established convergence properties may be fruitful to developing new optimization algorithms, primarily due to their physical interpretations, we have not pursued it here.

1.5 Nonconvex Optimization and Saddle-Point Escape

Unlike the convex setting where we can always achieve small objective gap $f(x) - \inf_{x \in \mathbb{R}^d} f(x) < \varepsilon$ for every $\varepsilon > 0$, this is intractable for non-convex functions in the worst case. Instead, the desired criteria becomes first- and second-order stationary points (FOSPs and SOSPs). When f is L -smooth and ρ -Hessian Lipschitz, the point $x \in \mathbb{R}^d$ is a first-order stationary point of $f(x)$ if $\|\nabla f(x)\| \leq \varepsilon$, and it is a second-order stationary point if $\lambda_{\min}(\nabla^2 f(x)) \geq -\sqrt{\rho\varepsilon}$. Locally around a second-order stationary point x , descent directions are governed by third-order terms. Furthermore, for many structured nonconvex

ML objectives, all second-order stationary points are (approximately) global minima (?). Higher order guarantees are relatively uncommon, while third-order stationary points can be found in polynomial time under reasonable smoothness assumptions, finding a fourth-order local minima is NP-hard in general (?).

? introduced cubic-regularized Newton, which finds second-order stationary points using gradients and Hessians. Unfortunately, computing the full $d \times d$ Hessian is not practical for large ML models. So, there was a need for algorithms which can converge to SOSPs without accessing the full Hessian. ? show that, for functions satisfying the strict saddle condition, stochastic gradient descent converges to a local minimum in a polynomial number of iterations. However, the while the dependence on d is polynomial, it is substantially large.

? introduced the first (nearly) dimension free saddle escape optimization algorithm, by randomly perturbing the iterate upon reaching a FOSP. They show that, with high probability, sequences generated by gradient descent on two points initially separated by $O(\delta\varepsilon/\sqrt{d}L)$ cannot both get “stuck”, at least one must escape the saddle point. They demonstrate convergence to a second-order stationary point in a remarkable $O\left(\frac{L(f(x_0)-f^*)}{\varepsilon^2}\text{polylog}(d/\delta)\right)$ time. In subsequent work, ? were able to show that Accelerated Gradient Descent improves this convergence to $\tilde{O}\left(\frac{L^{1/2}\rho^{1/4}(f(x_0)-f^*)}{\varepsilon^{7/4}}\right)$.

Most importantly for us and the setting of Chapter ??, ? introduce the first zeroth-order algorithm with Gaussian search directions that can efficiently escape saddle points, complete with linear speedup in population size r when function evaluations are done in parallel, converging to a second-order stationary point in time

$$\tilde{O}\left(\frac{d \max\{L, L^2\} \rho^{3/2}(f(x_0) - f^*)}{\varepsilon^{5/2}r}\right)$$

Their paper provides a roadmap for the analysis subsequently carried out in Chapter ??.

However, the main technical lemma bounding the norm of the gradient estimator with high probability must be changed, which is where the majority of work in that chapter went to. While simply perturbing the algorithm in Chapter ?? could likely provide convergence to SOSP, we would prefer a zeroth-order algorithm with a sub-quadratic dependence on ε .

? propose an alternate approach than perturbing gradient descent. The basic idea is to accelerate when the function is almost convex, otherwise exploiting negative curvature. They design two subroutines, the first, called Almost-convex-AGD with convexity parameter $\gamma > 0$, which optimizes a function h satisfying $\nabla^2 h(x) \succeq -\gamma I$ using accelerated gradient descent repeatedly on γ -strongly convex surrogates of h . The second subroutine is negative-curvature descent. Using Hessian-vector products $\nabla^2 f(x)v$ (which can be done with access only to gradients), they apply Lanczos to find an approximately smallest eigenvector v . When $v^\top \nabla^2 f(x)v \leq -\frac{\alpha}{2}$ where α is a curvature threshold, they take a step in the direction of

$$-\frac{2|v^\top \nabla^2 f(x)v|}{\rho} \text{sgn}(v^\top \nabla f(x))v$$

to decrease f by at least $\alpha^3/12\rho^2$ (the sign is chosen so the linear term does not hurt this decrease). Thus this subroutine either guarantees objective progress or ensures the function is locally α -almost convex. In the case where it is locally α -almost convex, to ensure global almost-convexity, thereby allowing us to run the Almost-convex-AGD, they utilize the surrogate

$$f_k(x) = f(x) + L \left[\|x - x_k\| - \frac{\alpha}{\rho} \right]_+^2$$

which has zero penalty within distance α/ρ of x_k but ensures global 3α -almost convexity. Carefully choosing $\alpha \asymp \sqrt{\rho\varepsilon}$, their algorithm returns a SOSP in time

$$\tilde{O} \left(\frac{(f(x_0) - f^*)L^{1/2}\rho^{1/4}}{\varepsilon^{7/4}} \right)$$

Unlike [?], they achieve a sub-quadratic convergence in $1/\varepsilon$ to a second-order stationary point. The principal contribution of Chapter ?? is to provide the zeroth-order accelerated gradient descent subroutine needed to run almost-convex AGD in their paper.

Chapter 2

Preliminaries

In this chapter we provide some background in optimization, list the standard assumptions and definitions, and introduce the relevant probabilistic tools that will become central to our analysis in the sequel.

2.1 Optimization

The perfect optimization algorithm would take a function $f : \mathbb{R}^d \rightarrow \mathbb{R}$ and output a point $x^* \in \mathbb{R}^d$ satisfying

$$x^* \in \operatorname{argmin}_{x \in \mathbb{R}^d} f(x)$$

However, since deciding whether a quartic polynomial $p(x_1, \dots, x_d) \in \mathbb{R}[x_1, \dots, x_d]$ is globally non-negative,

$$p(x) \geq 0 \quad \text{for all } x \in \mathbb{R}^d$$

is NP-hard ?, it is proven in ? that even verifying whether a given point is a local minimum of a non-convex function is NP-hard in the worst case. So, the best we can hope for must

be more restrictive. For general non-convex functions, the right definition ends up being

Definition 2.1. A point $x \in \mathbb{R}^d$ is an ε -first order stationary point if

$$\|\nabla f(x)\| \leq \varepsilon$$

On the other hand, for convex f , we usually measure accuracy in terms of the function gap.

Definition 2.2. Let $x^* \in \operatorname{argmin}_{x \in \mathbb{R}^d} f(x)$. A point $x \in \mathbb{R}^d$ is ε -optimal if

$$f(x) - f(x^*) \leq \varepsilon$$

To reach a first-order stationary point, we need an idea of how much signal we get from the gradient versus noise. Recall that for any differentiable function $f(x)$,

$$f(x+h) = f(x) + \langle \nabla f(x), h \rangle + O(\|h\|^2)$$

The noise $O(\|h\|^2)$ term is quantified by the L -smooth assumption

Definition 2.3. A function $f \in \mathcal{C}^1(\mathbb{R}^d)$ is L -smooth if

$$\|\nabla f(x) - \nabla f(y)\| \leq L \|x - y\|$$

Which leads to the descent lemma, the starting point to the analysis of any gradient-based algorithm.

Lemma 2.4 (Descent Lemma). *If $f \in \mathcal{C}^1(\mathbb{R}^d)$ is L -smooth, then for any $h \in \mathbb{R}^d$,*

$$f(x+h) \leq f(x) + \langle \nabla f(x), h \rangle + \frac{L}{2} \|h\|^2$$

Proof. Writing $g(t) = f(x+th)$, we know that $\frac{d}{dt}g(t) = \langle \nabla f(x+th), h \rangle$, and therefore

$$g(1) - g(0) = \int_0^1 \frac{d}{dt}g(t)dt = \int_0^1 \langle \nabla f(x+th), h \rangle dt$$

Therefore,

$$f(x+h) - f(x) - \langle \nabla f(x), h \rangle = \int_0^1 \langle \nabla f(x+th) - \nabla f(x), h \rangle dt$$

By the Cauchy-Schwarz inequality and the L -smooth assumption,

$$\begin{aligned} |\langle \nabla f(x+th) - \nabla f(x), h \rangle| &\leq \|\nabla f(x+th) - \nabla f(x)\| \|h\| \\ &\leq tL \|h\|^2 \end{aligned}$$

Therefore,

$$|f(x+h) - f(x) - \langle \nabla f(x), h \rangle| \leq L \|h\|^2 \int_0^1 t dt = \frac{L}{2} \|h\|^2$$

□

The main optimization algorithm used in ML is gradient descent.

Algorithm 1 Gradient Descent

Require: Objective function $f : \mathbb{R}^d \rightarrow \mathbb{R}$, smoothness constant $L > 0$, initial point $x_0 \in \mathbb{R}^d$, number of iterations T

```

1: for  $t = 0, \dots, T-1$  do
2:    $x_{t+1} \leftarrow x_t - \frac{1}{L} \nabla f(x_t)$ 
3: end for
4: return  $x_T$ 

```

By using the L -smoothness assumption, gradient descent converges to a first-order stationary point in $O(L(f(x) - f^*)/\varepsilon^2)$ iterations where $f^* = f(x^*) = \operatorname{argmin}_{x \in \mathbb{R}^d} f(x)$. For convex functions, it demonstrates a linear convergence to an ε -optimal point in $O(L\|x - x^*\|^2/\varepsilon)$ iterations. Gradient descent is the starting point for all algorithms discussed in the sequel.

2.2 Zero-Order Oracles

The focus of this work will be on zero-order optimization algorithms. Simply put, we drop the assumption that we can access gradients $\nabla f(x)$. We can only evaluate the function (“oracle”) at points of our choosing $x \in \mathbb{R}^d$, $f(x)$. The simplest thing to do now is to calculate

$$\tilde{\nabla} f(x) = \sum_{i=1}^d \frac{f(x + \sigma e_i) - f(x - \sigma e_i)}{2\sigma} e_i$$

for any orthonormal basis $\{e_1, \dots, e_d\}$ of $\mathbb{R}^d$. We have

$$\left\| \tilde{\nabla} f(x) - \nabla f(x) \right\| \leq \frac{L\sigma\sqrt{d}}{2}$$

This is a simple approach applied successfully, but is often of little practical use: frontier LLMs have d in the trillions, and trillions of forward passes to calculate just one gradient is far worse than the usual backpropagation.

For a quadratic $f(x) = \frac{1}{2}x^\top Hx$ with symmetric H ,

$$\nabla f(x) = Hx = \left[\frac{f(x+e_1)-f(x-e_1)}{2} \quad \dots \quad \frac{f(x+e_d)-f(x-e_d)}{2} \right]^\top$$

Thus typically the gradient gives d times more information than any single forward pass $f(x + e_i)$. This suggests that any zero-order optimization algorithm for convergence to a first-order stationary point for general nonconvex f should take d times more queries in the worst case. To the best of our knowledge this lower bound is currently open. Nevertheless, we are unaware of any method in the literature achieving a sublinear d dependence for this problem.

Another way to use the oracle f is to query random perturbations. For a random direction $p \sim \mathcal{N}(0, I_d)$, and search radius σ , we can look at $f(x + \sigma p)$. For $p_1, \dots, p_r \stackrel{\text{i.i.d.}}{\sim} \mathcal{N}(0, I_d)$

where r is the population size, a Taylor expansion suggests the following estimator

$$\tilde{\nabla} f(x) = \frac{1}{r} \sum_{i=1}^r \frac{f(x + \sigma p_i) - f(x - \sigma p_i)}{2\sigma} p_i. \quad (2.1)$$

In the sequel, this is the only estimator for the gradient we use. Upon defining the gaussian density

$$\varphi_\sigma(z) = \frac{1}{(2\pi\sigma^2)^{d/2}} \exp\left(-\frac{\|z\|^2}{2}\right)$$

and the Gaussian smoothed objective

$$f_\sigma(x) = (f * \varphi_\sigma)(x) = \int_{\mathbb{R}^d} f(x+z) \varphi_\sigma(z) dz = \mathbb{E}_{p \sim \mathcal{N}(0, I_d)}[f(x + \sigma p)]$$

under standard integrability assumptions

$$\nabla f_\sigma(x) = \frac{1}{\sigma} \mathbb{E}[f(x + \sigma p)p]$$

which by symmetry of the Gaussian distribution yields,

$$\nabla f_\sigma(x) = \mathbb{E}[\tilde{\nabla} f(x)]$$

The advantage behind the central-difference version $\tilde{\nabla} f(x)$ and $\frac{1}{\sigma} f(x + \sigma p)p$ is that the first has good cancellation properties under a Taylor expansion

$$f(x + \sigma p) - f(x - \sigma p) = 2\sigma \langle \nabla f(x), p \rangle + O(\|\sigma p\|^3).$$

Since $\mathbb{E}[\tilde{\nabla} f(x)] = \nabla f_\sigma(x)$, the bias of the estimator is precisely the error introduced by Gaussian smoothing. The high-dimensional analysis of this bias was investigated in ?, where they proved

$$|\mathbb{E}[\tilde{\nabla} f(x)] - \nabla f(x)| = O_d(L\sigma\sqrt{d})$$

Thus choosing $\sigma = o(d^{-1/2})$ ensures this bias vanishes at infinity. The intuition is that f linearizes in high dimensions.

2.3 The Low-Rank Trick

In an ML model $\Phi_\theta(x)$, the parameters $\theta \in \mathbb{R}^{d \times m}$ are naturally thought of not as vectors but as matrices. While helpful that the estimator ?? lets us evaluate at only r perturbations per iteration rather than the full $d \times m$, since $p \in \mathbb{R}^{d \times m}$ has all Gaussian entries, every entry is non-zero almost surely. Thus evaluating $\Phi_{\theta+p}(x)$ is still slow as we need to first add the two matrices $\theta + p$ before evaluating, with memory $O(rd)$ just to store the perturbations.

The low rank trick, introduced in ?, is to sample $a \sim \mathcal{N}(0, I_d)$, $b \sim \mathcal{N}(0, I_m)$, and exploit $(\theta + ab^\top)x = \theta x + a(b^\top x)$. This reduces the memory to $O(r(d + m))$, which makes zero-order algorithms practical.

However, for independent gaussian $Z, W \sim \mathcal{N}(0, 1)$, the product ZW is no longer subgaussian, it is sub-exponential, meaning it has fat tails. This could, *a priori*, ruin the convergence of the algorithm, as $\mathbb{E}[\|\tilde{\nabla} f(x)\|^2]$ could grow too large. This is the central question of Chapter ??. To heed our analysis, for $A, B \in \mathbb{R}^{d \times m}$, we define the Frobenius inner product and the Frobenius norm,

$$\langle A, B \rangle_F = \text{Tr}(AB^\top) = \sum_{i,j} A_{i,j} B_{i,j}, \quad \|A\|_F = \sqrt{\sum_{i,j} A_{i,j}^2}$$

We define L -smoothness in the same way as before using this norm.

The Gaussian distribution is rotationally invariant, meaning for $p \sim \mathcal{N}(0, I)$, for any rotation matrix R satisfying $RR^\top = R^\top R = I$, we have $Rp \sim \mathcal{N}(0, I)$. Thus, we search all directions uniformly. However, $ab^\top$ does not have this property, as it is rank 1, we could design a rotation matrix which rotates it into M which is full rank. This raises the question, is rotational invariance necessary? What if we miss the direction of the gradient?

Definition 2.5. The random variable $X \in \mathbb{R}^d$ is isotropic if

$$\mathbb{E}[XX^\top] = I$$

this is equivalent to saying, for any $u \in \mathbb{R}^d$,

$$\mathbb{E}[\langle X, u \rangle^2] = \mathbb{E}[u^\top X X^\top u] = u^\top \mathbb{E}[X X^\top] u = \|u\|^2$$

While $ab^\top$ is not rotationally invariant, it is isotropic:

Lemma 2.6. *For $a \sim \mathcal{N}(0, I_d), b \sim \mathcal{N}(0, I_m)$ independent, for any $U \in \mathbb{R}^{d \times m}$,*

$$\mathbb{E}[\langle ab^\top, U \rangle_F^2] = \|U\|_F^2$$

Proof. We have

$$\mathbb{E}[\langle ab^\top, U \rangle_F^2] = \mathbb{E}[\text{Tr}(Uba^\top)^2]$$

By cyclic property of trace, we know that $\text{Tr}(ABC) = \text{Tr}(CAB)$, so the above equals

$$\mathbb{E}[(a^\top U b)^2] = \mathbb{E}[b^\top U^\top a a^\top U b]$$

By conditioning on b , using that a and b are independent, and that $a \sim \mathcal{N}(0, I)$,

$$\mathbb{E}[b^\top U^\top \mathbb{E}[U^\top a a^\top U \mid b] b] = \mathbb{E}[b^\top U^\top \mathbb{E}[a a^\top] U b] = \mathbb{E}[b^\top U^\top U b]$$

This equals

$$\mathbb{E}[\text{Tr}(U^\top U b b^\top)] = \text{Tr}(U^\top U \mathbb{E}[b b^\top]) = \text{Tr}(U^\top U) = \|U\|_F^2.$$

□

It will turn out that isotropy is exactly the condition we need for convergence to a first-order stationary point.

2.4 Probability Tools

Here we give a short background on measure-theoretic probability. We define a σ -algebra

Definition 2.7. For a state space Ω , $\mathcal{F} \subset 2^\Omega$ is a σ -algebra if:

1. $\Omega \in \mathcal{F}$
2. $\mathcal{F}$ is closed under complements: if $A \in \mathcal{F}$, then $A^c \in \mathcal{F}$
3. $\mathcal{F}$ is closed under countable unions: if $A_1, A_2, \dots \in \mathcal{F}$, then $A = A_1 \cup A_2 \cup \dots \in \mathcal{F}$.

By combining the last two properties $\mathcal{F}$ is also closed under countable intersections. $\mathcal{F}$ is the collection of *measurable sets*, or in probability, *events*.

Definition 2.8. $\mathbb{P} : \mathcal{F} \rightarrow [0, 1]$ is a probability measure if $\mathbb{P}(\emptyset) = 0$, $\mathbb{P}(\Omega) = 1$, and

$$\mathbb{P}\left(\bigcup_{i=0}^{\infty} E_i\right) = \sum_{i=0}^{\infty} \mathbb{P}(E_i)$$

The collection $(\Omega, \mathcal{F}, \mathbb{P})$ is called a probability space.

In stochastic optimization, we often want to talk about the information available at iteration t . Filtrations give us a natural way to do this.

Definition 2.9. $\{\mathcal{F}_t\}_{t=0}^{\infty}$ is a filtration on $(\Omega, \mathcal{F}, \mathbb{P})$ if each $\mathcal{F}_t \subset \mathcal{F}$ is σ -algebra, and

$$\mathcal{F}_0 \subset \mathcal{F}_1 \subset \mathcal{F}_2 \subset \dots$$

For Chapters ?? and ??, at each iteration t we sample $p_{t,1}, \dots, p_{t,r} \sim \mathcal{N}(0, I_d)$ to evaluate $\tilde{\nabla} f(x_t)$. Thus we define the following filtration

Definition 2.10. The zero-order filtration $\{\mathcal{F}_t\}_{t=0}^{\infty}$ is defined as

$$\mathcal{F}_t = \sigma(x_0, \dots, x_t, \{p_{\tau,i} : 0 \leq \tau \leq t-1, 1 \leq i \leq r\}).$$

which encapsulates all information available at time t .

Central to our analysis are concentration inequalities. Recall Hoeffding's inequality:

Lemma 2.11 (Hoeffding's inequality). *Let $X_1, \dots, X_n$ be independent random variables satisfying $a_i \leq X_i \leq b_i$ almost surely for every $1 \leq i \leq n$. Then, for every $t > 0$,*

$$\Pr\left(\left|\sum_{i=1}^n (X_i - \mathbb{E}[X_i])\right| \geq t\right) \leq 2 \exp\left(-\frac{2t^2}{\sum_{i=1}^n (b_i - a_i)^2}\right).$$

By re-arranging, we can get

Lemma 2.12. *Let $X_1, \dots, X_n$ be independent random variables satisfying $a_i \leq X_i \leq b_i$ almost surely for every $1 \leq i \leq n$. Then with probability at least $1 - \delta$, we have*

$$\left| \sum_{i=1}^n (X_i - \mathbb{E}[X_i]) \right| \leq \sqrt{\frac{1}{2} \sum_{i=1}^n (b_i - a_i)^2 \log(2/\delta)}$$

Using $a_i \leq X_i \leq b_i$, $\text{Var}(X_i) = \mathbb{E}[(X_i - \mathbb{E}(X_i))^2] \leq (b_i - a_i)^2$. A bernoulli random variable which is a_i and b_i with probability $1/2$ has variance $(b_i - a_i)^2/4$, so this is tight up to constant factors. Since X_i are independent, $\text{Var}(X_1 + \dots + X_n) = \text{Var}(X_1) + \dots + \text{Var}(X_n) = \sum_{i=1}^n (b_i - a_i)^2$. Hoeffding's inequality says that, when the $a_i \leq X_i \leq b_i$, the absolute value of the centered sum is not typically that much larger than its expectation.

However, this property is not limited to just $a_i \leq X_i \leq b_i$. It carries over to martingale difference sequences.

Definition 2.13 (Martingale Difference Sequence). $\{\xi_t\}_{t \geq 1}$ is a martingale difference sequence wrt filtration $\mathcal{F}_t$ if $\xi_t \in \mathcal{F}_t$ and

$$\mathbb{E}[\xi_t \mid \mathcal{F}_{t-1}] = 0$$

The main intuition is that in a martingale difference sequence, the variance of the full sum is the sum of the variance of each ξ_t , which is vastly superior than including all the cross covariance terms $\text{Cov}(\xi_i, \xi_j)$. This improves our bound substantially.

Lemma 2.14. *Define $S_T = \sum_{t=1}^T \xi_t$. Then*

$$\mathbb{E}[S_T^2] = \sum_{t=1}^T \mathbb{E}[\xi_t^2]$$

Proof. We have

$$\mathbb{E}[M_T^2] = \sum \mathbb{E}[\xi_t^2] + 2\mathbb{E}\left[\sum_{i < j} \xi_i \xi_j\right]$$

Now,

$$\mathbb{E} \left[\sum_{i < j} \xi_i \xi_j \right] = \sum_{i < j} \mathbb{E}[\mathbb{E}[\xi_i \xi_j \mid \mathcal{F}_i]] = \sum_{i < j} \mathbb{E}[\xi_i \mathbb{E}[\xi_j \mid \mathcal{F}_i]] = 0.$$

□

For us, these martingale terms ξ_i will always be subgaussian.

Definition 2.15. The random variable $X \in \mathbb{R}$ is σ -subgaussian if

$$\mathbb{E} \left[\exp \left(\frac{X^2}{3\sigma^2} \right) \right] \leq 2$$

Under this definition $\mathcal{N}(0, 1)$ is 1-subgaussian (the constant 3 is not standard). We have the following lemma which vastly generalizes Hoeffding's inequality and will help us substantially in the sequel,

Lemma 2.16 (Adapted Sub-Gaussian Hoeffding's inequality). *Let $\{\mathcal{F}_t\}_{t=0}^T$ be a filtration, and let X_t be $\mathcal{F}_t$ -measurable. Suppose that, conditional on $\mathcal{F}_{t-1}$, X_t is σ_t -subgaussian and mean zero. Then, for any $\delta \in (0, 1)$, with probability at least $1 - \delta$,*

$$\left| \sum_{t=1}^T X_t \right| \leq c \sqrt{\log \left(\frac{2}{\delta} \right) \sum_{t=1}^T \sigma_t^2},$$

where $c > 0$ is an absolute constant.

Convergence of stochastic gradient methods are standard in the literature, see for example ????. Concentration inequalities will allow us to upgrade that convergence in expectation to convergence guarantees that hold with high probability.

2.5 Notation

d and m will always be dimension constants, r will always be the population size, T will always be the iteration horizon, L will always be the smoothness constant, σ will always be

the smoothing radius, ε will always be the desired accuracy, and δ will always be the failure probability. We use $O(\cdot)$, $\Omega(\cdot)$, and $\Theta(\cdot)$ in the standard sense. The notation $\tilde{O}(\cdot)$ hides polylogarithmic factors in the problem parameters, including d , L , T , $f(x_0) - f^*$, ε and failure probability δ . Throughout the sequel, χ always denotes a polylogarithmic factor.

Chapter 3

Zeroth-Order Mirror Descent

Gradient descent iteratively takes a step in the gradient direction, gradually reducing function value. It successively builds better and better upper bounds on $f(x^*)$. However, it ignores the dual problem of finding lower bounds on $f(x^*)$. For convex functions, we have for all $u \in \mathbb{R}^d$,

$$f(u) \geq f(x) + \langle \nabla f(x), u - x \rangle$$

Thus, for some points $\{x_t\}_{t=0}^{T-1}$, we have

$$f(u) \geq \frac{1}{T} \sum_{t=0}^{T-1} f(x_t) + \frac{1}{T} \sum_{t=0}^{T-1} \langle \nabla f(x_t), u - x_t \rangle \quad (3.1)$$

Defining $\bar{x} = \frac{1}{T} \sum_{t=0}^{T-1} x_t$, convexity yields $f(\bar{x}) \leq \frac{1}{T} \sum_{t=0}^{T-1} f(x_t)$, and combining this with the above yields,

$$f(\bar{x}) - f(u) \leq \frac{1}{T} \sum_{t=0}^{T-1} \langle \nabla f(x_t), x_t - u \rangle =: R_T(u)$$

$R_T(u)$, using terminology from online learning, is referred to as the regret. Consider a regularized version

$$\tilde{R}_T(u) = \frac{1}{T} \left(-\frac{w(u)}{\alpha} + \sum_{t=0}^{T-1} \langle \nabla f(x_t), x_t - u \rangle \right)$$

where $\alpha > 0$ is a trade-off parameter and $w(u)$ is a strongly convex regularizer. By taking the updates $x_t = \operatorname{argmax}_u \tilde{R}_T(u)$, we will tune α to show that Mirror descent converges in $T = O(\rho^2/\varepsilon^2)$ iterations where ρ^2 is the average value of $\|\nabla f(x_t)\|^2$ across iterations. This is because intuitively, the upper bound (??) becomes better when $\frac{1}{T} \sum_{t=0}^{T-1} \|\nabla f(x_t)\|^2$ is small.

Assumption 3.1. f is ρ -Lipschitz continuous and convex over a closed convex set $Q \subset \mathbb{R}^d$.

Definition 3.2. $w : Q \rightarrow \mathbb{R}^d$ is a DGF if it is 1-strongly convex w.r.t. $\|\cdot\|$, in symbols, $w(y) \geq w(x) + \langle \nabla w(x), y - x \rangle + \frac{1}{2} \|y - x\|^2$. The Bregman divergence is defined

$$V_x(y) = w(y) - \langle \nabla w(y), y - x \rangle - w(x)$$

We have $V_x(x) = 0$ and $V_x(y) \geq \frac{1}{2} \|x - y\|^2 \geq 0$.

Classic examples are $w(x) = \frac{1}{2} \|x\|_2^2$ for the ℓ^2 norm which gives $V_x(y) = \frac{1}{2} \|y - x\|_2^2$, and $w(y) = \sum_i y_i \log(y_i)$ for the ℓ^1 norm with $Q = \Delta$, which gives KL divergence $V_x(y) = \sum_i y_i \log(y_i/x_i) \geq \frac{1}{2} \|x - y\|_1^2$.

3.1 Classic Mirror Descent

Definition 3.3 (Mirror Descent Step). The mirror descent step with step length α is

$$x^+ = \operatorname{Mirr}_x(\alpha \nabla f(x)) \quad \text{where} \quad \operatorname{Mirr}_x(\xi) = \operatorname{argmin}_{y \in Q} \{V_x(y) + \langle \xi, y - x \rangle\}.$$

Mirror descents core lemma is this:

Lemma 3.4 (?). *If $x_{t+1} = \text{Mirr}_{x_t}(\alpha \nabla f(x_t))$, then*

$$\forall u \in Q, \quad \alpha(f(x_t) - f(u)) \leq \alpha \langle \nabla f(x_t), x_t - u \rangle \leq \frac{\alpha^2}{2} \|\nabla f(x_t)\|^2 + V_{x_t}(u) - V_{x_{t+1}}(u).$$

Proof. Notice

$$\alpha \langle \nabla f(x_t), x_t - u \rangle = \langle \alpha \nabla f(x_t), x_t - x_{t+1} \rangle + \langle \alpha \nabla f(x_t), x_{t+1} - u \rangle$$

Since Q is convex and $x_{t+1} = \text{Mirr}_{x_t}(\alpha \nabla f(x_t))$ is minimal, it satisfies the first order condition

$$\langle \nabla V_{x_t}(x_{t+1}) + \alpha \nabla f(x_t), u - x_{t+1} \rangle \geq 0 \quad \forall u \in Q$$

Therefore the above is at most

$$\langle \alpha \nabla f(x_t), x_t - x_{t+1} \rangle + \langle -\nabla V_{x_t}(x_{t+1}), x_{t+1} - u \rangle$$

Using the three-point identity

$$\langle -\nabla V_x(y), y - u \rangle = V_x(u) - V_y(u) - V_x(y)$$

we get the above equals

$$\begin{aligned} & \langle \alpha \nabla f(x_t), x_t - x_{t+1} \rangle + V_{x_t}(u) - V_{x_{t+1}}(u) - V_{x_t}(x_{t+1}) \\ & \leq \left(\langle \alpha \nabla f(x_t), x_t - x_{t+1} \rangle - \frac{1}{2} \|x_t - x_{t+1}\|^2 \right) + (V_{x_t}(u) - V_{x_{t+1}}(u)) \\ & \leq \frac{\alpha^2}{2} \|\nabla f(x_t)\|^2 + (V_{x_t}(u) - V_{x_{t+1}}(u)) \end{aligned}$$

After recalling that $V_x(y) \geq \frac{1}{2} \|x - y\|^2$, using Cauchy-Schwarz $\langle a, b \rangle \leq \|a\| \cdot \|b\|$ and Young's inequality $2ab \leq a^2 + b^2$ in that order.

The first inequality follows because by convexity $f(u) \geq f(x_t) + \langle \nabla f(x_t), u - x_t \rangle$. $\square$

Using this, we can prove Mirror Descent converges.

Lemma 3.5 (?). *For $\bar{x} = \frac{1}{T} \sum_{t=0}^{T-1} f(x_t)$, if we choose $\alpha = \sqrt{2\Theta/\rho^2 T}$, we have*

$$f(\bar{x}) - f(x^*) \leq \rho \sqrt{\frac{2\Theta}{T}}$$

Proof. Telescoping, we get

$$\sum_{t=0}^{T-1} \alpha(f(x_t) - f(x^*)) \leq \sum_{t=0}^{T-1} \frac{\alpha^2}{2} \|\nabla f(x_t)\|^2 + V_{x_0}(x^*) - V_{x_T}(x^*)$$

Using convexity, $f(\bar{x}) \leq 1/T \sum_{t=0}^{T-1} f(x_t)$, and recalling that f is ρ -Lipschitz continuous, we have

$$f(\bar{x}) - f(x^*) \leq \frac{\alpha}{2} \rho^2 + \frac{1}{\alpha} \Theta,$$

where Θ is any number satisfying $\Theta \geq V_{x_0}(x^*)$. Now by choosing $\alpha = \sqrt{2\Theta/\rho^2 T}$ to balance the two terms, we are done. $\square$

3.2 Zero-Order Mirror Descent

In this section, we assume that the norm $\|\cdot\|$ is the ℓ^2 norm and $Q = \mathbb{R}^d$. We begin by introducing the zeroth-order gradient estimator.

Definition 3.6 (Zero-Order Estimator). The Zero-Order gradient estimator is defined as follows

$$\tilde{\nabla} f(x) = \frac{f(x + \sigma p) - f(x - \sigma p)}{2\sigma} \cdot p$$

where $p \sim \mathcal{N}(0, I_d)$.

The main result of this chapter is the following.

Theorem 3.7 (informal). *With iterates defined as $x_{t+1} = \text{Mirr}_{x_t}(\alpha \tilde{\nabla} f(x_t))$, where the perturbation sampled at time t is independent from the past, if f is quadratic and*

$$\alpha = \tilde{O}\left(\sqrt{\frac{\Theta}{\rho^2 T d}}\right)$$

if we let $\bar{x} = \frac{1}{T} \sum_{t=0}^{T-1} x_t$, with probability at least $1 - \delta$,

$$f(\bar{x}) - f^* \leq \tilde{O} \left(\sqrt{\frac{d\Theta\rho^2}{T}} \right).$$

This matches the convergence rate of Mirror descent presented in ?, up to the zero-order d penalty and log factors.

3.3 Regret Bound

Now, notice that the only place in the proof of Lemma ?? where we used that the update is evaluated at the true gradient $\nabla f(x_t)$ was to upper bound $f(x_t) - f(u)$. The gradient was not used in the proof, actually. So we get this lemma for free:

Lemma 3.8. *If $x_{t+1} = \text{Mirr}_{x_t}(\alpha \tilde{\nabla} f(x_t))$, then*

$$\forall u \in Q \quad \alpha \langle \tilde{\nabla} f(x_t), x_t - u \rangle \leq \frac{\alpha^2}{2} \left\| \tilde{\nabla} f(x_t) \right\|^2 + V_{x_t}(u) - V_{x_{t+1}}(u).$$

Notice now that for any $u \in Q$,

$$\alpha(f(x_t) - f(u)) \leq \alpha \langle \nabla f(x_t), x_t - u \rangle = \alpha \langle \tilde{\nabla} f(x_t), x_t - u \rangle + \alpha \langle \nabla f(x_t) - \tilde{\nabla} f(x_t), x_t - u \rangle$$

For simplicity of exposition in this chapter only we will assume there is no quadratic error and $\tilde{\nabla} f(x_t) = p_t p_t^\top \nabla f(x_t)$. Thus the above equals,

$$\begin{aligned} & \alpha \langle \tilde{\nabla} f(x_t), x_t - u \rangle + \alpha \langle (I - p_t p_t^\top) \nabla f(x_t), x_t - u \rangle \\ & \leq \frac{\alpha^2}{2} \left\| \tilde{\nabla} f(x_t) \right\|^2 + V_{x_t}(u) - V_{x_{t+1}}(u) + \alpha \langle (I - p_t p_t^\top) \nabla f(x_t), x_t - u \rangle \end{aligned} \quad (3.2)$$

3.4 Concentration Bounds

Lemma 3.9. *With probability at least $1 - \delta$, we have for every $1 \leq t \leq T$,*

$$\left\| p_t p_t^\top \nabla f(x_t) \right\| \leq c\sqrt{d} \log(cT/\delta) \left\| \nabla f(x_t) \right\|$$

Proof. By Lemma ??, who's proof is deferred to Chapter ??, the event

$$A_t = \left\{ \|p_t\| \leq \sqrt{3d \log(C/\delta)} \right\}$$

has probability at least $1 - \delta$. Now, as $p_t^\top \nabla f(x_t) \sim \mathcal{N}(0, \|\nabla f(x_t)\|^2)$ conditional on $\mathcal{F}_t$, we know that

$$\mathbb{E} \left[\exp \left(\frac{\|p_t p_t^\top \nabla f(x_t) \mathbb{1}_A\|^2}{t^2} \right) \right] \leq \mathbb{E} \left[\exp \left(\frac{3d \log(C/\delta) |p_t^\top \nabla f(x_t)|^2}{t^2} \right) \right]$$

Using $\mathbb{E}[e^{\lambda Z^2}] = (1 - 2\lambda\sigma^2)^{-1/2}$ for $Z \sim \mathcal{N}(0, \sigma^2)$, we know that

$$\mathbb{E} \left[\exp \left(\frac{3d \log(C/\delta) |p_t^\top \nabla f(x_t)|^2}{t^2} \right) \mid \mathcal{F}_t \right] = \left(1 - \frac{6d \log(C/\delta) \|\nabla f(x_t)\|^2}{t^2} \right)^{-1/2}$$

The RHS is ≤ 2 for $t = \sqrt{8d \log(C/\delta)} \|\nabla f(x_t)\|$. By taking expectations on both sides this bound also holds unconditionally. Then using Markov's inequality,

$$\begin{aligned} & \mathbb{P} \left(\left\| p_t p_t^\top \nabla f(x_t) \mathbb{1}_A \right\| \geq \sqrt{8d \log(C/\delta)} \|\nabla f(x_t)\| \lambda \right) \\ &= \mathbb{P} \left(\exp \left(\frac{\|p_t p_t^\top \nabla f(x_t) \mathbb{1}_A\|^2}{8d \log(C/\delta) \|\nabla f(x_t)\|^2} \right) \geq e^{\lambda^2} \right) \leq 2e^{-\lambda^2} \end{aligned}$$

Now choosing $\lambda = \sqrt{\log(C/\delta)}$, possibly changing C to ensure $C \geq 2$, we have, with probability at least $1 - \delta$ on A ,

$$\left\| p_t p_t^\top \nabla f(x_t) \right\| \leq \log(C/\delta) \sqrt{8d} \|\nabla f(x_t)\|$$

Since A has probability at least $1 - \delta$, this bound holds with probability at least $1 - 2\delta$.

Rescaling $\delta \leftarrow \delta/2$ and changing the absolute constant yields the lemma. $\square$

This lemma highlights the main technique for creating concentration bounds when the underlying distribution has fat tails. The first thing you do is truncate with high probability, and prove the resulting random variable has similar scale (whether that be sub-gaussian, sub-exponential, etc.) as what you want, up to log factors.

The main idea behind the rest of this proof is that the mean-0 martingale error

$$\sum_{t=0}^{T-1} \alpha \langle (I - p_k p_k^\top) \nabla f(x_k), x_k - u \rangle$$

can hopefully be bounded by the other terms of the equation. In this case, it ends up being the “energy” term $V_{x_t}(u)$ and the Lipschitz constant ρ .

The main lemma which is different from the normal mirror descent bounds is the following.

Lemma 3.10. *Let $\{\mathcal{F}_t\}$ be a filtration. Assume that $g_t, d_t \in \mathcal{F}_{t-1}$, and assume $p_t \sim \mathcal{N}(0, I)$ with $p_t \in \mathcal{F}_t$, $p_t \perp \mathcal{F}_{t-1}$. Define*

$$E_\tau = \sum_{t=0}^{\tau-1} g_t^\top (I - p_t p_t^\top) d_t$$

Then for any θ , $0 < \delta < 1/e$, with probability at least $1 - \delta$,

$$|E_\tau| \leq c\theta \sum_{t=0}^{\tau-1} \log\left(\frac{c\tau}{\delta}\right) \|g_t\|^2 \|d_t\|^2 + \frac{1}{\theta} \log\left(\frac{c\tau}{\delta}\right).$$

Proof. For each t let

$$A_t = \left\{ L_\delta \leq |p_t^\top d_t| \leq \sqrt{3 \log(c/\delta) \|d_t\|^2} \right\}, \quad \mathbb{P}_{\mathcal{F}_{t-1}}(A_t) \geq 1 - \delta$$

Then by a lemma from before we know that $\mathbb{E}_{\mathcal{F}_{t-1}}[(I - p_t p_t^\top) d_t; A_t] = 0$. In particular,

$$\mathbb{E}_{\mathcal{F}_{t-1}}[g_t^\top (I - p_t p_t^\top) d_t; A_t] = 0.$$

Dropping the subscript t for simplicity,

$$\mathbb{E} \left[\exp \left(\frac{(g^\top p p^\top d \mathbb{1}_A)^2}{t^2} \right) \right] \leq \mathbb{E} \left[\exp \left(\frac{3 \log(C/\delta) \|d\|^2 (p^\top g)^2}{t^2} \right) \right]$$

Now, $p^\top g$ is $\mathcal{N}(0, \|g\|^2)$ random variable. Therefore, $\mathbb{E} \left[\exp \left((p^\top g)^2 / 4 \|g\|^2 \right) \right] \leq 2$. Now, taking $c \geq 1$ and $0 < \delta \leq 1/e$, we have

$$(g^\top (I - p p^\top) d)^2 \leq 2(g^\top d)^2 + 2(g^\top p p^\top d)^2$$

So, the random variable $Q_t = g_t^\top (I - p_t p_t^\top) d_t \mathbb{1}_{A_t}$ is $24 \|g_t\|^2 \|d_t\|^2 \log(C/\delta)$ subgaussian and mean 0 given $\mathcal{F}_t$. Thus by the (subgaussian) Hoeffding inequality with probability at least $1 - \delta$ for any θ ,

$$\left| \sum_{t=0}^{\tau-1} Q_t \right| \leq c\theta \sum_{t=0}^{\tau-1} \log(c/\delta) \|g_t\|^2 \|d_t\|^2 + \frac{1}{\theta} \log\left(\frac{c}{\delta}\right)$$

Now $\sum_{t=0}^{\tau-1} Q_t = E_\tau$ on $\bigcap A_t$. Notice, since $\mathbb{P}(A_t) \geq 1 - \delta$, $\mathbb{P}(\bigcap A_t) \geq 1 - \tau\delta$. Therefore with probability at least $1 - (\tau + 1)\delta$, E_τ satisfies the same bound as above. Rescaling δ to $\delta/(\tau + 1)$ and we have completed the proof. $\square$

This above lemma is the key for making the argument work. Because $V_x(y) \geq \frac{1}{2} \|x - y\|^2$, and recalling that f is ρ -Lipschitz continuous, we have

$$\left| \sum_{t=0}^{\tau-1} \alpha \langle (I - p_t p_t^\top) \nabla f(x_t), x_t - u \rangle \right| \leq \alpha \sqrt{c \sum_{t=0}^{\tau-1} \log^2(c\tau/\delta) \rho^2 V_{x_t}(u)}$$

after rescaling c and by choosing the optimal θ according to AM-GM $(x + y)/2 \geq \sqrt{xy}$. So, it remains to bound $\sum V_{x_t}(u)$. We can do this because Lemma ?? actually tells us a bound on $V_{x_\tau}(x^*)$.

3.5 Energy Bound

So, by telescoping Equation (??), we arrive at the following lemma.

Lemma 3.11. *Define $\chi = \log(cT^2/\delta)$. Then with probability at least $1 - \delta$, for every $1 \leq \tau \leq T$,*

$$V_{x_\tau}(x^*) \leq \frac{c\tau d\chi^2 \alpha^2}{2} \rho^2 + V_{x_0}(x^*) + \alpha\rho \sqrt{c \sum_{t=0}^{\tau-1} \chi^2 V_{x_t}(x^*)}$$

Proof. Telescoping Equation (??), we get

$$\begin{aligned} \sum_{t=0}^{\tau-1} \alpha(f(x_t) - f(x^*)) &\leq \sum_{t=0}^{\tau-1} \frac{\alpha^2}{2} \left\| \tilde{\nabla} f(x_t) \right\|^2 + V_{x_0}(x^*) - V_{x_\tau}(x^*) \\ &\quad + \sum_{t=0}^{\tau-1} \alpha \langle (I - p_t p_t^\top) \nabla f(x_t), x_t - x^* \rangle \end{aligned}$$

Therefore, by the previous discussion, by Lemma ?? and using that $f(x_t) \geq f(x^*)$ for any x_t ,

$$V_{x_\tau}(x^*) \leq \frac{c\tau d\chi^2\alpha^2}{2}\rho^2 + V_{x_0}(x^*) + \alpha\rho\sqrt{c\sum_{t=0}^{\tau-1} \log^2(c\tau/\delta)V_{x_t}(x^*)}$$

We can have this hold jointly for all $1 \leq \tau \leq T$ by rescaling $\delta \leftarrow \delta/T$. Then by using $\tau \leq T$, we arrive at the statement of the lemma. $\square$

Naturally, this above lemma should tell us how much $V_{x_t}(x^*)$ can grow. This next lemma tells us an upper bound on how much that is.

Lemma 3.12. *Let Θ be any upper bound on $V_{x_0}(x^*)$. Then for some absolute constant c ,*

$$\sup_{0 \leq t \leq T} V_{x_t}(x^*) \leq 2\Theta + cd\alpha^2\rho^2\chi^2T$$

Proof. Define $M = \sup_{0 \leq t \leq T} V_{x_t}(x^*)$. Notice that the right hand side is increasing in τ . Therefore applying sup to both sides, we get

$$\begin{aligned} M &\leq \frac{cTd\chi^2\alpha^2}{2}\rho^2 + \Theta + \alpha\rho\sqrt{c\sum_{t=0}^{T-1} \chi^2 V_{x_t}(x^*)} \\ &\leq \frac{cTd\chi^2\alpha^2}{2}\rho^2 + \Theta + \alpha\rho\sqrt{cT\chi^2 M} \end{aligned}$$

Using Young's inequality, we know that

$$\alpha\rho\sqrt{cT\chi^2 M} \leq \frac{1}{2}M + c\alpha^2\rho^2\chi^2T$$

Therefore,

$$\begin{aligned} M &\leq cTd\chi^2\alpha^2\rho^2 + 2\Theta + 2c\alpha^2\rho^2\chi^2T \\ &\leq 2\Theta + cd\alpha^2\rho^2\chi^2T \end{aligned}$$

for some absolute constant c as desired. $\square$

A natural question is: Is this a good bound at all? Shouldn't x_t be getting closer to the minimizer x^* ? But we showed $V_{x_t}(x^*)$ grows linearly?

To answer this, recall from before that $\alpha = \sqrt{2\Theta/\rho^2T}$. Ignoring the polylog factor $\chi^{3/2}$, and dimension d , we have actually shown that $V_{x_t}(x^*)$ stays bounded. But still, shouldn't it be decreasing to 0? Actually, it should not, since f could have many minimizers. The best bound you could hope to satisfy for every f is that it is just bounded.

For a similar choice of α , this tells us the accumulated error

$$\sum_{t=0}^{T-1} \alpha \langle (I - p_t p_t^\top) \nabla f(x_t), x_t - x^* \rangle$$

is around the same size as $V_{x_0}(x^*)$. This observation is what let's us prove convergence of Zero-Order mirror descent.

3.6 Putting it all Together

Theorem 3.13 (Zero-Order Mirror Descent Convergence). *Let $\bar{x} = 1/T \sum_{t=0}^{T-1} x_t$, if we choose*

$$\alpha = \sqrt{\frac{2\Theta}{\rho^2 c \chi^2 T d}}$$

then with probability at least $1 - \delta$,

$$f(\bar{x}) - f(x^*) \leq \frac{1}{T\alpha} \cdot 9\Theta = \tilde{O}\left(\rho\sqrt{\frac{\Theta d}{T}}\right)$$

Proof. By telescoping and using $V_{x_T}(x^*) \geq 0$, we get

$$\sum_{t=0}^{T-1} \alpha(f(x_t) - f^*) \leq \frac{cTd\chi^2\alpha^2}{2}\rho^2 + \Theta + \alpha\sqrt{c\sum_{t=0}^{T-1}\chi^2\rho^2V_{x_t}(x^*)}$$

By using the previous lemma, we know that

$$\alpha\sqrt{c\sum_{t=0}^{T-1}\chi^2\rho^2V_{x_t}(x^*)} \leq \alpha\sqrt{cT\chi^2\rho^2(2\Theta + c\alpha^2\rho^2\chi^2T)}$$

By choosing $\alpha = \sqrt{2\Theta/\rho^2c\chi^2Td} \leq \sqrt{2\Theta/\rho^2c\chi^2T}$ as $d \geq 1$, we know that

$$c\alpha^2\rho^2\chi^2T \leq 2\Theta$$

and therefore,

$$\alpha\sqrt{cT\chi^2\rho^2(2\Theta + c\alpha^2\rho^2\chi^2T)} \leq 4\Theta$$

Now again by our choice of α ,

$$\frac{cTd\chi^2\alpha^2}{2}\rho^2 \leq 4\Theta$$

Therefore,

$$\alpha(f(\bar{x}) - f^*) \leq \frac{9\Theta}{T}$$

We conclude,

$$f(\bar{x}) - f^* \leq \frac{9\Theta}{T\alpha} = \rho c \sqrt{\frac{9}{2}} \sqrt{\frac{d\Theta}{T}} \chi = \tilde{O}\left(\rho \sqrt{\frac{d\Theta}{T}}\right).$$

□

The extension of this analysis to the setting without quadratic error is left to future work. The main purpose of this chapter was not merely to show that mirror descent converges, but to determine whether zeroth-order mirror descent can be analyzed with the zeroth-order gradient estimator. Having answered this question affirmatively, the techniques developed here will be applied extensively in the next chapter.

Chapter 4

Accelerated Zeroth-Order Gradient Descent via Linear Mixing

Accelerated gradient descent for L -smooth convex functions was developed in ?, improving the iteration complexity of finding an ε -optimal point from $O(1/\varepsilon)$ with gradient descent to $O(1/\sqrt{\varepsilon})$. The algorithm can be defined inductively as

$$\begin{aligned}x_k &= y_{k-1} - s\nabla f(y_{k-1}) \\y_k &= x_k + \frac{k-1}{k+2}(x_k - x_{k-1})\end{aligned}$$

The point y_k “looks ahead” in the direction of velocity $x_k - x_{k-1}$ with coefficient $(k-1)/(k+2) \approx 1 - 3/k$, then x_{k+1} subsequently does a gradient step at that point. This is slightly different from the momentum method

$$\begin{aligned}x_{k+1} &= y_k - s\nabla f(x_k) \\y_k &= x_k + \beta(x_k - x_{k-1})\end{aligned}$$

which exhibits acceleration for strongly convex quadratics, but may not even converge for general convex f . It is a mystery why the precise coefficient $(k-1)/(k+2)$ was chosen by Nesterov, except as an algebraic trick to make an energy telescope. Recently, by taking a limit as the step size $s \rightarrow 0$, [?](#) find that Nesterov's AGD yields the second-order scheme

$$\ddot{X}_t + \frac{3}{t}\dot{X}_t + \nabla f(X_t) = 0$$

This is unusual not least because Nesterov's AGD is a first-order method, only relying on gradients. Perhaps surprisingly, their analysis found that $3 = 2 + 1$ the smallest constant which preserves acceleration. Unfortunately, these Lyapunov proofs prove difficult to analyze with high probability, as martingale fluctuations yield an additive term, so the energy could grow linearly rather than decreasing.

To our rescue is the linear mixing framework discovered by [?](#). By “mixing” mirror and gradient descent,

$$\begin{aligned} x_{k+1} &= \lambda z_k + (1 - \lambda)y_k \\ y_k &= x_k - \frac{1}{L}\nabla f(x_k) \\ z_k &= \text{Mirr}_{z_{k-1}}(\alpha \nabla f(x_k)) \end{aligned}$$

after carefully tuning α and λ , [?](#) show this scheme can also achieve acceleration. The point x_{k+1} is a linear combination of y_k and z_k , y_k does a gradient step at x_k , and z_k does a mirror step with $\nabla f(x_k)$. The idea is that, while mirror and gradient descent tell us to go to different places, we can use part of both answers to achieve a faster scheme than regular gradient descent. Most importantly for us, they do not rely on carefully engineered Lyapunov functions, and the analysis is relatively direct. This is what will allow us to design a zero-order accelerated gradient descent scheme.

4.1 The Main Result

Assumption 4.1. We assume that the smoothness constant of f is $L \geq 1$. Otherwise, we can replace it with 1. Also assume that $0 < \delta < 1/e$.

We use the linear mixing framework of ? to design an accelerated zero-order gradient method. At each iteration, the gradient sequence (y_t) and the mirror sequence (z_t) are mixed to form a common query point. The same zeroth-order estimate at this point is then used for both updates. The full procedure is given in Algorithm ??.

Definition 4.2 (Rank- r Symmetric Zeroth-Order Estimator). For $x \in \mathbb{R}^d$ and independent directions $p_1, \dots, p_r \sim \mathcal{N}(0, I_d)$, define

$$\tilde{\nabla} f(x) := \frac{1}{r} \sum_{i=1}^r \frac{f(x + \sigma p_i) - f(x - \sigma p_i)}{2\sigma} p_i.$$

with each iteration using $2r$ function evaluations.

Algorithm 2 Rank- r Zeroth-Order Linear Mixing

Require: Zeroth-order oracle f and initial states $x_0, y_0, z_0 \in \mathbb{R}^d$

Require: Horizon T , rank r , stepsizes $\eta, \alpha > 0$, smoothing radius $\sigma > 0$, and mixing weight $\lambda \in (0, 1)$

for $t = 0, \dots, T - 1$ **do**

 Set $x_{t+1} \leftarrow \lambda z_t + (1 - \lambda)y_t$

 Sample $p_{t+1,1}, \dots, p_{t+1,r} \stackrel{\text{i.i.d.}}{\sim} \mathcal{N}(0, I_d)$ to compute $\tilde{\nabla} f(x_{t+1})$

 Update $y_{t+1} \leftarrow x_{t+1} - \eta \tilde{\nabla} f(x_{t+1})$

 Update $z_{t+1} \leftarrow \text{Mirr}_{z_t}(\alpha \tilde{\nabla} f(x_{t+1}))$

end for

return $\bar{x}_T \leftarrow \frac{1}{T} \sum_{t=1}^T x_t$

The main result of this chapter is the following (informal) theorem.

Theorem 4.3 (informal). *For a given initial point x_0 satisfying $f(x_0) - f^* \leq \Delta$ and letting*

$\Theta \geq V_{x_0}(x^*)$ be any upper bound, by choosing

$$\eta = \tilde{O}\left(\frac{r}{dL}\right), \quad \alpha = \tilde{O}\left(\frac{r}{d}\sqrt{\frac{\Theta}{L\Delta}}\right), \quad \sigma^2 \leq \tilde{O}\left(\frac{r^3\Delta^{3/2}}{L^{7/2}d^6\sqrt{\Theta}}\right),$$

$$\frac{1-\lambda}{\lambda} = \tilde{O}\left(\frac{d^2L\Theta}{\Delta}\right), \quad r = \tilde{O}(1)$$

Algorithm ?? outputs a point $\bar{x}$ satisfying $f(\bar{x}) - f^* \leq \Delta/2$ in

$$T = \tilde{O}\left(\frac{d}{r}\sqrt{\frac{L\Theta}{\Delta}}\right)$$

iterations.

Furthermore, by restarting Algorithm ?? $\log_2(\Delta/\varepsilon)$ times, we receive a point $\bar{x}$ satisfying $f(\bar{x}) - f^* \leq \varepsilon$ in

$$T = \tilde{O}\left(\frac{d}{r}\sqrt{\frac{L\Theta}{\varepsilon}}\right)$$

iterations.

4.2 Quadratic Error

The first thing to do is to control the quadratic error in this estimator.

Lemma 4.4 (Quadratic Error).

$$\left\| \tilde{\nabla} f(x) - \frac{1}{r} \sum_{i=1}^r p_i p_i^\top \nabla f(x) \right\| \leq L\sigma \frac{1}{r} \sum_{i=1}^r \|p_i\|^3$$

Proof. We have

$$\begin{aligned} \frac{f(x + \sigma p) - f(x - \sigma p)}{2\sigma} p - pp^\top \nabla f(x) &= \frac{1}{2\sigma} \left[\int_{-1}^1 \frac{d}{dt} f(x + t\sigma p) - \sigma pp^\top \nabla f(x) dt \right] \\ &= \frac{1}{2\sigma} \left[\int_{-1}^1 \langle \nabla f(x + t\sigma p) - \nabla f(x), \sigma p \rangle p dt \right] \end{aligned}$$

Now, by L -smoothness,

$$\left| \int_{-1}^1 \langle \nabla f(x + \sigma t p) - \nabla f(x), \sigma p \rangle p dt \right| \leq \int_{-1}^1 L\sigma^2 |t| \|p\|^3 dt$$

Therefore,

$$\left| \frac{f(x + \sigma p) - f(x - \sigma p)}{2\sigma} p - pp^\top \nabla f(x) \right| \leq L\sigma \|p\|^3$$

Then by triangle inequality,

$$\left\| \tilde{\nabla} f(x) - \frac{1}{r} \sum_{i=1}^r p_i p_i^\top \nabla f(x) \right\| \leq L\sigma \frac{1}{r} \sum_{i=1}^r \|p_i\|^3.$$

□

Unfortunately, this bound is necessary. Ideally we would like to just say that $f(x + \sigma p) - f(x - \sigma p)/2\sigma p = pp^\top \nabla g(x)$ for some function $g(x)$ related to $f(x)$. Possibly a smoothed variant f_σ . But this cannot possible hold for every p unless f is quadratic in a neighborhood of x . To see this, we can use MVT to expand the LHS as $\langle \nabla f(x + \theta\sigma p), p \rangle p$. By taking $p = e_i$, we see that $e_i^\top \nabla g(x) = e_i^\top \nabla f(x + \theta_i \sigma e_i)$. Sending $\sigma \rightarrow 0$, $\nabla f(x + \theta_i \sigma e_i) \rightarrow \nabla f(x)$ under L -smoothness assumption, meaning that $\nabla g(x) = \nabla f(x)$. This would say that f is quadratic, which is not true in general.

4.3 Gradient Progress

These lemmas tell us how much progress the gradient descent step is making.

Lemma 4.5 (? Corollary 7.3.2). *Let $P \in \mathbb{R}^{d \times r}$ be a matrix with i.i.d. Gaussian entries.*

Define $\|A\|_{\text{op}} = \sup_{u \in \mathbb{S}^{r-1}} \|Au\|_2$. Then

$$\mathbb{P}\left(\|P\|_{\text{op}} \geq \sqrt{d} + \sqrt{r} + s\right) \leq 2e^{-cs^2}$$

Proposition 4.6. *Define $\chi = \log(CrT/\delta)$ and let $P_t = \begin{bmatrix} p_{t,1} & \cdots & p_{t,r} \end{bmatrix}$. Define the bounded operator norm event*

$$\mathfrak{D}_t = \left\{ \frac{1}{\sqrt{r}} \|P\|_{\text{op}} \leq C \frac{d}{r} \sqrt{\log(cTr/\delta)} \right\}$$

Then on $\mathfrak{D}_t$, for every $0 \leq t \leq T-1$,

$$\left\| \frac{1}{r} \sum_{i=1}^r p_{t,i} p_{t,i}^\top \nabla f(x_t) \right\|^2 \leq \frac{Cd}{r} \log(cTr/\delta) \cdot \frac{1}{r} \sum_{i=1}^r |p_{t,i}^\top \nabla f(x_t)|^2$$

Furthermore,

$$\mathbb{P} \left(\bigcap_{t=0}^{T-1} \mathfrak{D}_t \right) \geq 1 - \delta$$

Proof. Write $P = \begin{bmatrix} p_{t,1} & \cdots & p_{t,r} \end{bmatrix} \in \mathbb{R}^{d \times r}$. By choosing $s = \sqrt{1/c \log(2/\delta)}$ above, we know that

$$\frac{1}{\sqrt{r}} \|P\|_{\text{op}} \leq 1 + \sqrt{\frac{d}{r}} + \frac{1}{c\sqrt{r}} \sqrt{\log(c/\delta)} \leq C \sqrt{\frac{d}{r}} \sqrt{\log(c/\delta)}$$

with probability at least $1 - \delta$. As $PP^\top = \sum_{i=1}^r p_{t,i} p_{t,i}^\top$, we see that

$$\left\| \frac{1}{r} PP^\top \nabla f(x_t) \right\|^2 \leq \left(\frac{1}{\sqrt{r}} \|P\|_{\text{op}} \right)^2 \frac{1}{r} \left\| P^\top \nabla f(x_t) \right\|^2 \leq \frac{Cd}{r} \log(c/\delta) \frac{1}{r} \sum_{i=1}^r |p_{t,i}^\top \nabla f(x_t)|^2$$

We then rescale δ to δ/Tr to complete the proof. $\square$

Lemma 4.7. *Define the big progress event*

$$\mathfrak{P}_t = \left\{ \frac{1}{r} \sum_{i=1}^r |\langle p_{t,i}, \nabla f(x_t) \rangle|^2 \geq \frac{1}{2} \|\nabla f(x_t)\|^2 \right\}$$

As long as $r \geq 16\chi$, we have

$$\mathbb{P} \left(\bigcap_t \mathfrak{P}_t \right) \geq 1 - \delta.$$

Proof. As $\langle p_{t,i}, \nabla f(x_t) \rangle \stackrel{i.i.d.}{\sim} \mathcal{N}(0, \|\nabla f(x_t)\|^2)$, by the Laurent-Massart inequality ([?], Lemma 1, Equation (4.4)),

$$\mathbb{P} \left(\chi_r^2 \leq \frac{r}{2} \right) \leq \exp \left(-\frac{r}{16} \right)$$

Therefore as long as $r \geq 16\chi \geq 16 \log(T/\delta)$,

$$\frac{1}{r} \sum_{i=1}^r |\langle p_{t,i}, \nabla f(x_t) \rangle|^2 \geq \frac{1}{2} \|\nabla f(x_t)\|^2$$

for all $0 \leq t \leq T - 1$. $\square$

Discussion. It is unfortunate to necessitate a lower bound on r which will eventually grow with ε . The way around this is usually to instead look at the gradient progress over a time interval I of length $\asymp \log(T/\delta)$, in which case we could apply the same bound as above. However, this would require the gradients to be of roughly the same size, otherwise one gradient could dominate and we would only really have r effective samples even over the whole interval (compared to $r \log(T/\delta)$).

When proving gradient descent converges to a first-order stationary point (in the sequel), this is possible, and alleviates this requirement on r . However, in the mixing case, since the z_t point depends only on z_{t-1} and not on y_{t-1} , the mirrored sequence could drift extremely far from the gradient sequence. In this case, $x_{t+1} - x_t = (1 - \lambda)(y_t - x_t) + \lambda z_t - \lambda x_t$ could potentially grow unbounded. For example, consider the one-variable function $f(x) = Gx + \frac{L}{2}x^2$. Assume that $x_t = 0$, we know that $\nabla f(x_t) = G$. Then $y_t = -\eta G$. $z_t = z_{t-1} - \alpha G$. Then

$$|x_{t+1} - x_t| = |x_{t+1}| = |(1 - \lambda) - \eta G + \lambda(z_{t-1} - \alpha G)|$$

Sending $|z_{t-1}| \rightarrow \infty$, this gap can become unbounded. The α dictating z_{t-1} 's progression grows with ε , so this is (a priori) not an unrealistic scenario. Furthermore, we will restart the algorithm later, with $z_0 \neq x_0$, so z_0 really could be far away from y_0 .

Lemma 4.8. *Define the bounded-perturbation event*

$$\mathfrak{B}_{t+1} := \bigcap_{i=1}^r \left\{ \sqrt{dL_{\delta/rT}} \leq \|p_{t+1,i}\| \leq \sqrt{3d\chi} \right\}$$

Then on $\mathfrak{B}_{t+1} \cap \mathfrak{P}_{t+1}$, as long as $\eta \leq r/(12Ld\chi)$, we have

$$f(y_{t+1}) \leq f(x_{t+1}) - \frac{\eta}{8} \|\nabla f(x_{t+1})\|^2 + C' L^2 \sigma^2 \chi^3 d^3$$

Furthermore,

$$\mathbb{P}\left(\bigcap_{t=0}^{T-1} \mathfrak{B}_t\right) \geq 1 - \delta$$

Proof. By L -smoothness,

$$\begin{aligned} f(y_{t+1}) &\leq f(x_{t+1}) + \langle \nabla f(x_{t+1}), y_{t+1} - x_{t+1} \rangle + \frac{L}{2} \|y_{t+1} - x_{t+1}\|^2 \\ &= f(x_{t+1}) - \eta \langle \nabla f(x_{t+1}), \tilde{\nabla} f(x_{t+1}) \rangle + \frac{L\eta^2}{2} \|\tilde{\nabla} f(x_{t+1})\|^2 \end{aligned}$$

Writing $\tilde{\nabla} f(x_{t+1}) = \frac{1}{r} \sum_{i=1}^r p_i p_i^\top \nabla f(x_{t+1}) + R_{t+1}$, we get

$$\begin{aligned} f(y_{t+1}) &\leq f(x_{t+1}) - \frac{\eta}{r} \sum_{i=1}^r |p_i^\top \nabla f(x_{t+1})|^2 - \eta \langle \nabla f(x_{t+1}), R_{t+1} \rangle \\ &\quad + L\eta^2 \left\| \frac{1}{r} \sum_{i=1}^r p_i p_i^\top \nabla f(x_{t+1}) \right\|^2 + L\eta^2 \|R_{t+1}\|^2 \end{aligned}$$

Now, $2\langle \eta \nabla f(x_{t+1}), R_{t+1} \rangle \leq \eta^2 \|\nabla f(x_{t+1})\|^2 + R_{t+1}^2$ by Cauchy-Schwarz and Young's inequality.

Define the event where p_i are bounded:

$$\mathfrak{B}_{t+1} := \bigcap_{i=1}^r \left\{ \sqrt{dL\delta/rT} \leq \|p_{t+1,i}\| \leq \sqrt{3d\chi} \right\}$$

We know that $\mathbb{P}(\mathfrak{B}_{t+1}) \geq 1 - \delta/rT$ by Lemma ???. Thus,

$$\begin{aligned} f(y_{t+1}) &\leq f(x_{t+1}) - \frac{\eta}{r} \sum_{i=1}^r |p_i^\top \nabla f(x_{t+1})|^2 + \eta^2 \gamma \|\nabla f(x_{t+1})\|^2 + (L\eta^2 + \gamma) \|R_{t+1}\|^2 \\ &\quad + \frac{CLd\eta^2 \log(Tr/\delta)}{r} \cdot \frac{1}{r} \sum_{i=1}^r |p_i^\top \nabla f(x_{t+1})|^2 \end{aligned}$$

From quadratic error lemma we know that $|R_{t+1}| \leq 3L\sigma \log^{3/2}(1/\delta)d^{3/2}$. Now by choosing $\eta \leq r/(4CLd\chi)$, we know that $-\eta + CLd\chi\eta^2/r \leq -3/4\eta$. As $L\eta^2 < 1$ as $L \geq 1$, possibly changing the absolute constant C' , we get,

$$f(y_{t+1}) \leq f(x_{t+1}) - \frac{3\eta}{4} \frac{1}{r} \sum_{i=1}^r |p_i^\top \nabla f(x_{t+1})|^2 + \eta^2 \|\nabla f(x_{t+1})\|^2 + C'L^2\sigma^2\chi^3d^3$$

As long as $\eta \leq 1/8$, $-3\eta/8 + \eta^2 \leq -\eta/4$, so using the previous lemma on $\mathfrak{P}_{t+1}$, we ensure

$$f(y_{t+1}) \leq f(x_{t+1}) - \frac{\eta}{8} \|\nabla f(x_{t+1})\|^2 + C' L^2 \sigma^2 \chi^3 d^3$$

□

4.4 Mirror Descent against Progress

We can copy the mirror descent lemma from last time:

Lemma 4.9. *If $z_{t+1} = \text{Mirr}_{z_t}(\alpha \tilde{\nabla} f(x_{t+1}))$, then*

$$\forall u \in \mathbb{R}^d \quad \alpha \langle \tilde{\nabla} f(x_{t+1}), z_t - u \rangle \leq \frac{\alpha^2}{2} \left\| \tilde{\nabla} f(x_{t+1}) \right\|^2 + V_{z_t}(u) - V_{z_{t+1}}(u).$$

The standard coupling lemma comes from using $\|\nabla f(x_{t+1})\|^2 \leq 2L(f(x_{t+1}) - f(y_{t+1}))$.

However this doesn't work here, as we are using an estimator. But we can fix that using the following lemma.

Lemma 4.10. *Define the event*

$$\mathfrak{E}_{t+1} = \left\{ \left\| \frac{1}{r} \sum_{i=1}^r p_{t+1,i} p_{t+1,i}^\top \nabla f(x_{t+1}) \right\|^2 \leq \frac{Cd}{r} \chi \|\nabla f(x_{t+1})\|^2 \right\}$$

On $\mathfrak{B}_{t+1} \cap \mathfrak{E}_{t+1} \cap \mathfrak{D}_{t+1} \cap \mathfrak{P}_{t+1}$, we have for all slack $\gamma \geq 2$,

$$\frac{Cd\gamma\chi}{r} \|\nabla f(x_{t+1})\|^2 + \left\| \tilde{\nabla} f(x_{t+1}) \right\|^2 \leq \frac{16Cd\gamma\chi}{r\eta} (f(x_{t+1}) - f(y_{t+1}) + C' L^2 \sigma^2 \chi^3 d^3)$$

Furthermore,

$$\mathbb{P}\left(\bigcap \mathfrak{E}_t\right) \geq 1 - \delta.$$

Proof. Define the event

$$\mathfrak{E}_{t+1} = \left\{ \left\| \frac{1}{r} \sum_{i=1}^r p_{t+1,i} p_{t+1,i}^\top \nabla f(x_{t+1}) \right\|^2 \leq \frac{Cd}{r} \chi \|\nabla f(x_{t+1})\|^2 \right\}$$

Then on $\mathfrak{G}_{t+1}$,

$$\begin{aligned} \left\| \tilde{\nabla} f(x_{t+1}) \right\|^2 &\leq 2 \left\| \frac{1}{r} \sum_{i=1}^r p_i p_i^\top \nabla f(x_{t+1}) \right\|^2 + 2 \|R_{t+1}\|^2 \\ &\leq 2 \frac{Cd}{r} \chi \cdot \|\nabla f(x_{t+1})\|^2 + 2 \|R_{t+1}\|^2 \\ &\leq \frac{2Cd}{r} \chi \|\nabla f(x_{t+1})\|^2 + C' L^2 \sigma^2 \log^3(rT/\delta) d^3 \end{aligned}$$

We will need some slack of the gradient norm to tune later. For $\gamma \geq 2$, as $\gamma - \gamma/2 \geq 1$, this is at most

$$\frac{2Cd\gamma}{r} \chi \|\nabla f(x_{t+1})\|^2 + C' L^2 \sigma^2 \log^3(rT/\delta) d^3 - \frac{Cd\gamma}{r} \chi \|\nabla f(x_{t+1})\|^2$$

From before we know that

$$\begin{aligned} &\frac{\eta}{8} \|\nabla f(x_{t+1})\|^2 \\ &\leq f(x_{t+1}) - f(y_{t+1}) + C' L^2 \sigma^2 \log^3(rT/\delta) d^3 \end{aligned}$$

Therefore an upper bound on $\left\| \tilde{\nabla} f(x_{t+1}) \right\|^2$ is

$$\begin{aligned} &\frac{16Cd\gamma\chi}{r\eta} [f(x_{t+1}) - f(y_{t+1}) + C' L^2 \sigma^2 \log^3(rT/\delta) d^3] \\ &+ C' L^2 \sigma^2 \chi^3 d^3 - \frac{Cd\gamma}{r} \chi \|\nabla f(x_{t+1})\|^2 \end{aligned}$$

As $\eta < 1/8$ we can absorb the last term into the first by changing the constant C' . Finally, after an application of the triangle inequality and Lemma ??, we have $\mathbb{P}(\mathfrak{G}_{t+1}) \geq 1 - \delta/T$, which gives the statement of the lemma. $\square$

4.5 Linear Mixing

This lemma explains how to choose the mixing parameter λ .

Lemma 4.11. When λ is chosen so $\frac{1-\lambda}{\lambda} = \frac{8Cd\gamma\chi\alpha}{\eta}$, the “regret” is bounded by

$$\alpha\langle\nabla f(x_{t+1}), x_{t+1} - u\rangle - \alpha\langle\nabla f(x_{t+1}), z_t - u\rangle \leq \frac{8Cd\gamma\chi\alpha^2}{\eta}(f(y_t) - f(x_{t+1}))$$

Proof. We have by convexity and the definition of $x_{t+1} = \lambda z_t + (1-\lambda)y_t$,

$$\begin{aligned} & \alpha\langle\nabla f(x_{t+1}), x_{t+1} - u\rangle - \alpha\langle\nabla f(x_{t+1}), z_t - u\rangle \\ &= \alpha\langle\nabla f(x_{t+1}), x_{t+1} - z_t\rangle = \frac{(1-\lambda)\alpha}{\lambda}\langle\nabla f(x_{t+1}), y_t - x_{t+1}\rangle \\ &\leq \frac{(1-\lambda)\alpha}{\lambda}(f(y_t) - f(x_{t+1})) \end{aligned}$$

Now let $\lambda \in (0, 1)$ satisfy $\frac{1-\lambda}{\lambda} = \frac{8Cd\gamma\chi\alpha}{\eta}$ and we are done. $\square$

This lemma reveals the martingale term we need to bound.

Lemma 4.12.

$$\begin{aligned} \alpha\langle\nabla f(x_{t+1}), z_t - u\rangle &\leq \alpha\langle\tilde{\nabla} f(x_{t+1}), z_t - u\rangle + \alpha\sum_{i=1}^r\langle(I - p_i p_i^\top)\nabla f(x_{t+1}), z_t - u\rangle \\ &\quad + C'T\alpha^2\|R_{t+1}\|^2 + \frac{1}{4T}V_{z_t}(u). \end{aligned}$$

Proof.

$$\langle\nabla f(x_{t+1}), z_t - u\rangle = \langle\tilde{\nabla} f(x_{t+1}), z_t - u\rangle + \sum_{i=1}^r\langle(I - p_i p_i^\top)\nabla f(x_{t+1}), z_t - u\rangle + \langle R_{t+1}, z_t - u\rangle$$

Now, by Young’s inequality, $\alpha\langle R_{t+1}, z_t - u\rangle \leq C'\alpha^2 T\|R_{t+1}\|^2 + \frac{1}{8T}\|z_t - u\|^2$, by strong convexity of Bregman divergence, $V_{z_t}(u) \geq \frac{1}{2}\|z_t - u\|^2$, so we arrive at the statement of the lemma. $\square$

4.6 Dealing with the Martingale

The main idea from here is that, the martingale term

$$\sum_{t=0}^{T-1} \sum_{i=1}^r \langle(I - p_i p_i^\top)\nabla f(x_{t+1}), z_t - u\rangle$$

can be bounded by the “Energy” $V_{z_t}(x^*)$, which itself can be bounded by the sum of the gradient norms $\sum \|\nabla f(x_t)\|^2$ and $\Theta \geq V_{z_0}(x^*)$.

This highlights a very general principle. To deal with martingale error terms with high probability, try bounding them by some other part of the equation which should have typical size $\Theta(1)$. In this case, that will be $V_{z_t}(x^*)$ for us.

Lemma 4.13. *Define*

$$\mathfrak{M}_{t+1} = \left\{ \left| \sum_{t=0}^{\tau-1} \sum_{i=1}^r \alpha \langle (I - p_i p_i^\top) \nabla f(x_{t+1}), z_t - u \rangle \right| \leq \alpha \sqrt{Cr \sum_{t=0}^{\tau-1} \chi^2 \|\nabla f(x_t)\|^2 V_{z_t}(u)} \right\}$$

Then

$$\mathbb{P} \left(\bigcap_{t=0}^{T-1} \mathfrak{M}_t \right) \geq 1 - \delta.$$

Proof. From Lemma ?? above, we know that

$$\begin{aligned} & \left| \sum_{t=0}^{\tau-1} \sum_{i=1}^r \alpha \langle (I - p_i p_i^\top) \nabla f(x_{t+1}), z_t - u \rangle \right| \\ & \leq C\theta \sum_{t=0}^{\tau-1} \sum_{i=1}^r \log(CrT/\delta) \|\nabla f(x_{t+1})\|^2 \|z_t - u\|^2 + \frac{1}{\theta} \log\left(\frac{CrT}{\delta}\right) \end{aligned}$$

Using $V_{z_t}(u) \geq \frac{1}{2} \|z_t - u\|^2$ and balancing the two terms by carefully choosing θ according to AM-GM $2\sqrt{ab} \leq a + b$, we arrive at the statement of the lemma. $\square$

4.7 Putting it all Together

First, as in the proof of Mirror Descent, we bound the regret, then $V_{z_t}(x^*)$, then last we will tune α .

Lemma 4.14 (Regret Bound). *On $\bigcap_{t=1}^T (\mathfrak{B}_t \cap \mathfrak{D}_t \cap \mathfrak{P}_t \cap \mathfrak{G}_t \cap \mathfrak{M}_t)$, we have*

$$\begin{aligned}
& \sum_{t=0}^{\tau-1} \alpha \langle \nabla f(x_t), x_t - x^* \rangle \\
& \leq \frac{8Cd\gamma\chi}{r\eta} \alpha^2 (f(y_0) - f(y_\tau)) - \frac{C\alpha^2 d\gamma\chi}{2r} \sum_{t=0}^{\tau-1} \|\nabla f(x_{t+1})\|^2 \\
& + \alpha \sqrt{\frac{C}{r} \sum_{t=0}^{\tau-1} \chi^2 \|\nabla f(x_{t+1})\|^2 V_{z_t}(x^*)} + \frac{C'(\alpha^2 T + \alpha T^2) \gamma L^2 \sigma^2 \chi^4 d^4}{r\eta} \\
& + V_{z_0}(x^*) - V_{z_\tau}(x^*) + \frac{1}{4T} \sum_{t=0}^{\tau-1} V_{z_t}(x^*)
\end{aligned}$$

Proof. By Lemma ??, we know that

$$\sum_{t=0}^{\tau-1} \alpha \langle \nabla f(x_t), x_t - x^* \rangle \leq \sum_{t=0}^{\tau-1} \frac{8Cd\gamma\chi\alpha^2}{\eta} (f(y_t) - f(x_{t+1})) + \sum_{t=0}^{\tau-1} \alpha \langle \nabla f(x_{t+1}), z_t - u \rangle$$

By Lemma ??, we know that

$$\begin{aligned}
& \sum_{t=0}^{\tau-1} \alpha \langle \nabla f(x_{t+1}), z_t - u \rangle \\
& \leq \alpha \sum_{t=0}^{\tau-1} \left[\langle \tilde{\nabla} f(x_{t+1}), z_t - u \rangle + \frac{1}{r} \sum_{i=1}^r \langle (I - p_i p_i^\top) \nabla f(x_{t+1}), z_t - u \rangle + 2T \|R_{t+1}\|^2 + \frac{1}{T} V_{z_t}(u) \right]
\end{aligned}$$

From above we know that

$$\alpha \sum_{t=0}^{\tau-1} \sum_{i=1}^r \langle (I - p_i p_i^\top) \nabla f(x_{t+1}), z_t - u \rangle \leq \alpha \sqrt{Cr \sum_{t=0}^{\tau-1} \chi^2 \|\nabla f(x_{t+1})\|^2 V_{z_t}(u)}$$

From Lemmas ?? and ?? we know that

$$\begin{aligned}
& \sum_{t=0}^{\tau-1} \alpha \langle \tilde{\nabla} f(x_{t+1}), z_t - u \rangle \\
& \leq \alpha^2 \sum_{t=0}^{\tau-1} \frac{8Cd\gamma\chi}{r\eta} (f(x_{t+1}) - f(y_{t+1}) + C' L^2 \sigma^2 \chi^3 d^3) - \frac{C\alpha^2 d\gamma\chi}{2r} \sum_{t=0}^{\tau-1} \|\nabla f(x_{t+1})\|^2 \\
& + V_{z_0}(x^*) - V_{z_\tau}(x^*)
\end{aligned}$$

Recall that $\|R_{t+1}\|^2 \leq C'L^2\sigma^2\chi^3d^3$. Putting all of it together, we get:

$$\begin{aligned}
& \sum_{t=0}^{\tau-1} \alpha \langle \nabla f(x_t), x_t - x^* \rangle \\
& \leq \frac{8Cd\gamma\chi}{r\eta} \alpha^2 (f(y_0) - f(y_\tau)) - \frac{C\alpha^2 d\gamma\chi}{2r} \sum_{t=0}^{\tau-1} \|\nabla f(x_{t+1})\|^2 \\
& + \alpha \sqrt{\frac{C}{r} \sum_{t=0}^{\tau-1} \chi^2 \|\nabla f(x_{t+1})\|^2 V_{z_t}(x^*)} + \frac{C'\alpha^2 T \gamma L^2 \sigma^2 \chi^4 d^4}{r\eta} + C'\alpha T^2 L^2 \sigma^2 \chi^3 d^3 \\
& + V_{z_0}(x^*) - V_{z_\tau}(x^*) + \frac{1}{4T} \sum_{t=0}^{\tau-1} V_{z_t}(x^*)
\end{aligned}$$

By using $r \leq d$ and $\eta < 1$ to handle the two quadratic terms we arrive at the statement of the lemma. $\square$

This next lemma is what we meant by bounding the martingale by other already-present $\Theta(1)$ terms.

Lemma 4.15 (Energy Bound). *On $\bigcap_{t=1}^T (\mathfrak{B}_t \cap \mathfrak{D}_t \cap \mathfrak{P}_t \cap \mathfrak{G}_t \cap \mathfrak{M}_t)$, upon defining $M = \sup_{0 \leq t \leq T-1} V_{z_t}(x^*)$, choosing the slack $\gamma = 8\chi$ and letting $\Theta \geq V_{z_0}(x^*)$ be any upper bound, we have*

$$\frac{M}{2} \leq \frac{256Cd\chi^2}{r\eta} \alpha^2 (f(y_0) - f^*) + \Theta + \frac{C'(\alpha^2 T + \alpha T^2) L^2 \sigma^2 \chi^4 d^4}{r\eta}$$

Proof. By convexity we know that $f^* = f(x^*) \geq f(x) + \langle \nabla f(x), x^* - x \rangle$. Rearranging, we get $\langle \nabla f(x), x^* - x \rangle \geq f(x) - f^* \geq 0$. Using Young's inequality, we know that

$$\alpha \sqrt{\frac{C}{r} \chi^2 \sum_{t=0}^{\tau-1} \|\nabla f(x_{t+1})\|^2 V_{z_t}(x^*)} \leq \frac{M}{4} + \alpha^2 \frac{4C\chi^2}{r} \sum_{t=0}^{\tau-1} \|\nabla f(x_{t+1})\|^2$$

Therefore by the previous lemma, and using $\frac{1}{T} \sum_{t=0}^{T-1} V_{z_t}(x^*) \leq \frac{1}{4}M$, we know that

$$\begin{aligned} V_{z_\tau}(x^*) &\leq \frac{8C\gamma d\chi}{r\eta} \alpha^2 (f(y_0) - f(y_\tau)) - \frac{C\alpha^2 d\gamma\chi}{2r} \sum_{t=0}^{\tau-1} \|\nabla f(x_{t+1})\|^2 \\ &\quad + V_{z_0}(x^*) + \frac{M}{4} + \frac{C'(\alpha^2 T + \alpha T^2) \gamma L^2 \sigma^2 \chi^4 d^4}{r\eta} \\ &\quad + \alpha^2 \frac{4C\chi}{r} \sum_{t=0}^{\tau-1} \|\nabla f(x_{t+1})\|^2 + \frac{M}{4} \end{aligned}$$

By choosing $\gamma = 8\chi$, as $d \geq 1$, we know that

$$-\frac{C\alpha^2 d\gamma\chi}{2r} \sum_{t=0}^{\tau-1} \|\nabla f(x_{t+1})\|^2 + \alpha^2 \frac{4C\chi^2}{r} \sum_{t=0}^{\tau-1} \|\nabla f(x_{t+1})\|^2 \leq 0$$

By using $f(y_0) - f(y_\tau) \leq f(y_0) - f^*$ along with $V_{z_0}(x^*) \leq \Theta$, and taking supremum of both sides, we have

$$M \leq \frac{64Cd\chi^2}{r\eta} \alpha^2 (f(y_0) - f^*) + \Theta + \frac{M}{2} + \frac{C'(\alpha^2 T + \alpha T^2) L^2 \sigma^2 \chi^4 d^4}{r\eta}$$

which completes the proof. $\square$

Theorem 4.16 (Halving Function Objective). *Assuming the initial error $f(x_0) - f^* \leq \Delta$, and defining $\bar{x} = 1/T \sum_{t=0}^{T-1} x_t$, by choosing*

$$\eta = \frac{r}{4CLd\chi}, \quad \alpha = \frac{r}{16Cd\chi^{3/2}} \sqrt{\frac{\Theta}{L\Delta}}, \quad \sigma^2 \leq \frac{r^3 \Delta^{3/2}}{4C' L^{7/2} \chi^{15/2} d^6 \sqrt{\Theta}}, \quad r \geq 16\chi$$

In

$$T = \frac{256Cd\chi^{3/2}}{r} \sqrt{\frac{L\Theta}{\Delta}}$$

iterations we have $f(\bar{x}) - f^* \leq \Delta/2$ with probability at least $1 - 5\delta$.

Proof. On $\bigcap_{t=1}^T (\mathfrak{B}_t \cap \mathfrak{D}_t \cap \mathfrak{P}_t \cap \mathfrak{G}_t \cap \mathfrak{M}_t)$ all our previous lemmas hold. Each has probability at least $1 - \delta$ so the intersection of 5 has probability at least $1 - 5\delta$.

By Lemma ??, we know that

$$\begin{aligned}
& \sum_{t=0}^{T-1} \alpha \langle \nabla f(x_t), x_t - x^* \rangle \\
& \leq \frac{64Cd\chi}{r\eta} \alpha^2 (f(y_0) - f(y_T)) - \frac{C\alpha^2 d\gamma\chi}{2r} \sum_{t=0}^{T-1} \|\nabla f(x_{t+1})\|^2 \\
& + \alpha \sqrt{\frac{C}{r} \sum_{t=0}^{T-1} \chi^2 \|\nabla f(x_{t+1})\|^2} V_{z_t}(x^*) + \frac{C'(\alpha^2 T + \alpha T^2) L^2 \sigma^2 \chi^4 d^4}{r\eta} \\
& + V_{z_0}(x^*) - V_{z_T}(x^*) + \frac{1}{4T} \sum_{t=0}^{T-1} V_{z_t}(x^*)
\end{aligned}$$

Using Young's inequality in exactly the way as the previous lemma, we can rearrange this to

$$\begin{aligned}
& \sum_{t=0}^{T-1} \alpha \langle \nabla f(x_t), x_t - x^* \rangle \\
& \leq \frac{64Cd\chi^2}{r\eta} \alpha^2 (f(y_0) - f^*) + \frac{C'(\alpha^2 T + \alpha T^2) L^2 \sigma^2 \chi^4 d^4}{r\eta} + \Theta + M/4
\end{aligned}$$

Using the previous lemma, we have

$$\begin{aligned}
& \sum_{t=0}^{T-1} \alpha \langle \nabla f(x_t), x_t - x^* \rangle \\
& \leq \frac{128Cd\chi^2}{r\eta} \alpha^2 (f(y_0) - f^*) + \frac{C'(\alpha^2 T + \alpha T^2) L^2 \sigma^2 \chi^4 d^4}{r\eta} + 2\Theta
\end{aligned}$$

Let $\bar{x} = \sum_{t=0}^{T-1} x_t$. By convexity,

$$T(f(\bar{x}) - f^*) \leq \sum_{t=0}^{T-1} (f(x_t) - f^*) \leq \sum_{t=0}^{T-1} \langle \nabla f(x_t), x_t - x^* \rangle$$

Therefore, assuming the initial objective gap $f(y_0) - f^* \leq \Delta$,

$$f(\bar{x}) - f^* \leq \frac{128Cd\chi^2\alpha\Delta}{r\eta T} + \frac{2\Theta}{\alpha T} + \frac{C'(\alpha + T)L^2\sigma^2\chi^4d^4}{r\eta}$$

Now choose the learning rate $\eta = r/(4CLd\chi)$. Then we get

$$f(\bar{x}) - f^* \leq \frac{512C^2d^2L\chi^3\alpha\Delta}{r^2T} + \frac{2\Theta}{\alpha T} + \frac{C'(\alpha + T)L^3\sigma^2\chi^5d^5}{r^2}$$

By choosing

$$\alpha = \frac{r}{16Cd\chi^{3/2}} \sqrt{\frac{\Theta}{L\Delta}},$$

and

$$T = \frac{256Cd\chi^{3/2}}{r} \sqrt{\frac{L\Theta}{\Delta}}$$

as $L \geq 1$ we have $T \geq \alpha$, so the above is at most

$$\frac{\Delta}{4} + \sqrt{\frac{L\Theta}{\Delta}} \frac{C'L^3\sigma^2\chi^{15/2}d^6}{r^3}$$

Then as long as

$$\sigma^2 \leq \frac{r^3\Delta^{3/2}}{4C'L^{7/2}\chi^{15/2}d^6\sqrt{\Theta}}$$

We have

$$f(\bar{x}) - f^* \leq \frac{\Delta}{2}$$

thereby halving the objective value. This completes the proof. $\square$

We want to restart this halving procedure $\log(1/\varepsilon)$ times so we can converge to a point with objective gap $f(\bar{x}) - f^* \leq \varepsilon$. However, the $\bar{x}$ returned might not satisfy $V_{\bar{x}}(x^*) \leq \Theta$. Indeed, this valuation could actually go up if there is multiple minima. This point is actually unmentioned and unresolved in ?. The simple fix is to notice above that the requirement $z_0 = y_0$ is not necessary. Actually, the only fact we used about z_0 was that $V_{z_0}(x^*) \leq \Theta$. So by restarting the algorithm $n = \log_2(\Delta/\varepsilon)$ times with $x_0 = \bar{x} = y_0$, and z_0 as it's original

value each time, we can find a point with $f(\bar{x}) - f^* \leq \varepsilon$ in

$$T = \tilde{O} \left(\frac{d}{r} \sqrt{\frac{L\Theta}{\varepsilon}} + \frac{d}{r} \sqrt{\frac{L\Theta}{2\varepsilon}} + \frac{d}{r} \sqrt{\frac{L\Theta}{4\varepsilon}} + \cdots \right) = \tilde{O} \left(\frac{d}{r} \sqrt{\frac{L\Theta}{\Delta}} \right)$$

iterations, which yields the following theorem:

Theorem 4.17. *By restarting Algorithm ?? $\log_2(\Delta/\varepsilon)$ many times, we find a point $\bar{x}$ so that $f(\bar{x}) - f^* \leq \varepsilon$ in*

$$T = \tilde{O} \left(\frac{d}{r} \sqrt{\frac{L\Theta}{\varepsilon}} \right)$$

iterations.

When $r = d$, this matches the asymptotically optimal running time to converge to an ε -optimal point for a L -smooth convex function, up to logarithmic factors. To the best of our knowledge, this is the first zero-order accelerated gradient descent algorithm for smooth convex functions that achieves the optimal running time (up to log factors) which can be run once and output a solution with high probability. On the contrary, expectation-based methods need to run $\log(1/\delta)$ independent copies to ensure they work with high probability. We believe this analysis gives a strong intuition about the fluctuations one often sees when running the algorithm in practice, as we have directly compared the martingale term to the progress and distance to optimum.

Chapter 5

EGGROLL for Nonconvex Optimization

In previous chapters we have largely focused on the theoretical aspects of zero-order optimization methods. However, their implementation and practical use is also an important question. While we have focused on zero-order methods, how do large GPU clusters actually share the perturbations with each other? *A priori*, they would have to send the full weights of each perturbation to every cluster, which could take prohibitively long due to synchronization requirements. This would effectively eliminate the gains behind evaluating r perturbations in parallel we have discussed in detail above.

?? solved this problem by sharing only the pseudorandom seeds used to generate each Gaussian perturbation. Full high-dimensional perturbations can thus be shared between GPUs using only a 32-bit seed, where the receiving GPU subsequently reconstructs the perturbation locally. For the first time, zeroth-order methods, also called Evolutionary Strategies (ES), was scaled to machine learning models with millions of parameters, using more than one thousand parallel workers. But as we have seen in Chapter ??, increasing

the population size r linearly speeds up the algorithm. A two-times increase in the number of parallel workers yields precisely a two-times speedup in training speed.

The bottleneck now, having solved the problem of sharing the perturbations across GPUs, was memory. Each Gaussian perturbation has d nonzero entries almost surely, where d is the dimension of the model. Thus, for population size r , each GPU needs to store $O(d \cdot r)$ parameters. This fundamentally limits how big r can be in practice.

Very recently, ? introduced EGGROLL to solve this bottleneck by replacing full Gaussian perturbations $P \sim \mathcal{N}(0, I_{d \times m})$ with rank- r perturbations $A \in \mathbb{R}^{d \times k}$ and $B \in \mathbb{R}^{m \times k}$ with $E = \frac{1}{\sqrt{k}}AB^\top$, critically using $u(M + \sigma P)^\top = uM^\top + \frac{\sigma}{\sqrt{k}}(uB)A^\top$. Using this low-rank trick, they were able to scale ES vastly further than done previously. On the theoretical side, they showed that, under some reasonable assumptions if $\sigma = o(d^{-1/2})$ then as $d \rightarrow \infty$,

$$\frac{1}{\sigma} \mathbb{E}_{p \sim \mathcal{N}(0, I_d)} [f(x + \sigma P)P] \rightarrow \nabla f(x)$$

Furthermore, they prove that as the rank of the perturbations $k \rightarrow \infty$,

$$\frac{1}{\sigma} \mathbb{E}_{E \sim p(E)} [Ef(M + \sigma E)] \rightarrow \frac{1}{\sigma} \mathbb{E}_{p \sim \mathcal{N}(0, I_{d \times m})} [f(x + \sigma P)P]$$

this is because the distribution of E will approach $\mathcal{N}(0, I_{d \times m})$ with increasing rank.

5.1 Main Result

The preceding result suggests that increasing the rank of the perturbations is beneficial, so the resulting distribution becomes Gaussian. The Gaussian distribution is rotationally invariant and the rank-1 update $ab^\top$ is not, maybe we need this property to “locate” the gradient. However, this is not true, it turns out that the requirement for convergence is the isotropic condition mentioned previously in the preliminaries. We prove that, despite only using rank-1 perturbations $ab^\top$, not only do we get convergence to a first-order

stationary point, but that its speed also scales linearly with the population size. The practical takeaway is that when function evaluations are performed in parallel, twice as much compute converges twice as fast.

Assumption 5.1 (Properties of f). We suppose that $f : \mathbb{R}^{d \times m} \rightarrow \mathbb{R}$ satisfies the following properties:

1. f is differentiable and lower bounded, i.e. $f^* = \min_x f(x) > -\infty$
2. f is L -gradient Lipschitz, i.e.

$$\|\nabla f(X) - \nabla f(Y)\|_F \leq L \|X - Y\|_F \quad \forall X, Y \in \mathbb{R}^{d \times m}$$

Definition 5.2. A point $X \in \mathbb{R}^{d \times m}$ is an ε -first order stationary point if

$$\|\nabla f(X)\|_F \leq \varepsilon$$

In this excerpt we work with symmetric EGGROLL, following the low-rank perturbation framework of ?.

Definition 5.3 (Symmetric EGGROLL Update). We define a rank r symmetric zeroth order estimator as follows:

$$g_\sigma^r(X) := \frac{1}{r} \sum_{i=1}^r \frac{f(X + \sigma P_i) - f(X - \sigma P_i)}{2\sigma} P_i$$

using $P_i = a_i b_i^\top$ where $a_i \stackrel{i.i.d.}{\sim} \mathcal{N}(0, I_d)$ and $b_i \stackrel{i.i.d.}{\sim} \mathcal{N}(0, I_m)$, and $\sigma > 0$ is the smoothing radius.

Algorithm 3 Rank- r Symmetric EGGROLL

Require: Initial point $X_0 \in \mathbb{R}^{d \times m}$, horizon T , stepsize $\eta > 0$, smoothing radius $\sigma > 0$, rank $r \in \mathbb{N}$
for $t = 0, \dots, T - 1$ **do**
 Sample $\{a_{t,i}\}_{i=1}^r \sim \mathcal{N}(0, I)$ and $\{b_{t,i}\}_{i=1}^r \sim \mathcal{N}(0, I)$ to compute $g_\sigma^r(X_t)$
 Update $X_{t+1} \leftarrow X_t - \eta g_\sigma^r(X_t)$
end for
return X_T

We state an informal version of the main result.

Theorem 5.4 (informal). *Consider running Algorithm ?? . Suppose $0 < \delta \leq 1/e$. Suppose*

$$\sigma = \tilde{O}\left(\frac{\varepsilon}{dm}\right), \quad \eta = \tilde{O}\left(\frac{r}{dm}\right).$$

Then in

$$T = \tilde{\Omega}\left(\frac{dmL(f(X_0) - f^*)}{r\varepsilon^2}\right)$$

iterations (with each iteration using $2r$ function evaluations), with probability at least $1 - \delta$, at least $3/4$ of the iterates are ε -first order stationary points.

The overall proof strategy follows that of ?. The main new ingredient is Theorem ??, which establishes the concentration bound required when full Gaussian perturbations are replaced by rank-one perturbations. The Gaussian argument does not extend to this setting because rank-one perturbations have fat tails, with entries that are sub-exponential rather than sub-gaussian.

5.2 Concentration Bounds

Here we prove the novel concentration bound about the EGGROLL error. Throughout this excerpt, all norms of matrices is the Frobenius norm

$$\|A\|_F = \left(\sum_{i,j} A_{i,j}^2 \right)^{1/2}.$$

Throughout this paper, we denote

$$\mathbb{E}_{\mathcal{F}_{t-1}}[X; A] := \mathbb{E}[X \mathbb{I}_A \mid \mathcal{F}_{t-1}]$$

We have seen subGaussian random variables in the preliminaries. Here, we will need a generalization, called norm-subGaussian.

Definition 5.5. The random variable X is norm-subGaussian with parameter σ , abbreviated $\text{nSG}(\sigma)$, if

$$\mathbb{E} \left[\exp \left(\frac{\|X\|^2}{3\sigma^2} \right) \right] \leq 2$$

As an example, $p \sim \mathcal{N}(0, I_d)$ is $\text{nSG}(\sqrt{d})$. The next lemma is basically a Hoeffding-like inequality for norm-subGaussian random variables.

Lemma 5.6 (?). *Let random vectors $x_1, \dots, x_n \in \mathbb{R}^d$, and corresponding filtrations $\mathcal{F}_t = \sigma(X_1, \dots, X_t)$ satisfy that*

$$\mathbb{E}_{\mathcal{F}_{t-1}}[x_t] = 0, \quad \mathbb{P}_{\mathcal{F}_{t-1}}(\|x_t\| \geq s) \leq 2e^{-\frac{s^2}{3\sigma^2}} \quad \forall s \in \mathbb{R}$$

Then there exists an absolute constant c such for any fixed $\delta > 0, \theta > 0$, with probability at least $1 - \delta$,

$$\left\| \sum_{i=1}^n X_i \right\| \leq c \cdot \theta \sum_{i=1}^n \sigma_i^2 + \frac{1}{\theta} \log \left(\frac{2d}{\delta} \right).$$

This will help us later.

Lemma 5.7. *Let X follow χ_d^2 distribution and $0 < \delta < 1/e$. Then there exists L_δ such that for every d , with*

$$A := \{dL_\delta^2 \leq X \leq 3d \log(C/\delta)\}$$

we have

$$\mathbb{E}[(X - d); A] = 0, \quad \mathbb{P}(A) \geq 1 - \delta$$

for some absolute constant C .

Proof. Let $f_d(x)$ be the χ_d^2 density,

$$f_d(x) = \frac{1}{2^{d/2} \Gamma(d/2)} x^{d/2-1} e^{-x/2}, \quad x \geq 0.$$

A simple calculation shows that

$$(x - d)f_d(x) = -2 \frac{d}{dx} (x f_d(x))$$

Define $U = 3d \log(1/\delta)$ and define the set

$$A = \{L \leq X \leq U\}$$

We want

$$\mathbb{E}[(X - d); A] = \int_L^U (x - d)f_d(x)dx = -2L f_d(L) - 2U f_d(U) \stackrel{\text{want}}{=} 0.$$

Therefore, we need

$$\begin{aligned} L f_d(L) &= U f_d(U) \\ \iff L^d e^{-L} &= U^d e^{-U} \\ \iff U - L &= d \log(U/L) \end{aligned}$$

Now define $\ell = L/d$ and $u = U/d$. Then we want to solve

$$\begin{aligned} du - d\ell &= d \log(u/\ell) \\ \iff \ell e^{-\ell} &= u e^{-u} \end{aligned}$$

The function $g(x) = x e^{-x}$ is increasing on $[0, 1)$ and decreasing on $(1, \infty)$. Therefore, as long as $u > 1$, which follows as $\delta < 1/e$, there exists a unique $\ell < 1$ satisfying the above equation. Now notice, as $e^{-\ell} \geq e^{-1}$,

$$\ell e^{-1} \leq \ell e^{-\ell} = 3 \log(1/\delta) \delta^3 \leq 3 e^{-1} \delta^2$$

Now we have that

$$\mathbb{P}(A^c) = \mathbb{P}(X \geq U) + \mathbb{P}(X \leq L)$$

Recall the Chernoff bound For $0 < \alpha < 1$,

$$\mathbb{P}(X \leq \alpha d) \leq (\alpha e^{1-\alpha})^{d/2} \leq (e\alpha)^{d/2}$$

Then we have

$$\mathbb{P}(X \leq L) \leq \mathbb{P}(X \leq 3d\delta^2) \leq (3e\delta^2)^{d/2} = (\sqrt{3e}\delta)^d$$

When $\sqrt{3e}\delta < 1$, this is $< \sqrt{3e}\delta$, otherwise this bound holds vacuously. Therefore $\mathbb{P}(X \leq L) \leq \sqrt{3e}\delta$. On the other hand, using that for $t < 1/2$,

$$\mathbb{E}\left[e^{t\|Z\|^2}\right] = (1 - 2t)^{-d/2},$$

along with Markov's inequality, we get

$$\mathbb{P}(X \geq 3d \log(1/\delta)) = \mathbb{P}(\exp(X/3) \geq \exp(d \log(1/\delta))) \leq 3^{d/2} \delta^d \leq \sqrt{3}\delta$$

And thus $\mathbb{P}(A) \geq 1 - C\delta$ for the absolute constant $C = \sqrt{3} + \sqrt{3e}$. $\square$

Here is the novel concentration bound. The proof is inspired by Proposition 2 of ? but

with many important differences.

Theorem 5.8. *Let $\{\mathcal{F}_t\}_{t \geq -1}$ be a filtration, and let $\{a_t\}_{t \geq 0}, \{b_t\}_{t \geq 0}$ be two sequences of independent random vectors such that $a_t \sim \mathcal{N}(0, I_d)$, $b_t \sim \mathcal{N}(0, I_m)$, $a_t, b_t \in \mathcal{F}_t$ and a_t, b_t are independent of $\mathcal{F}_{t-1}$. Let $\{V_t\}_{t \geq 0}$ be a sequence of random matrices such that $V_t \in \mathcal{F}_{t-1}$. For each $\tau \geq 0$, let*

$$E_\tau = \sum_{t=0}^{\tau-1} \left(V_t - \langle V_t, a_t b_t^\top \rangle a_t b_t^\top \right)$$

Then, there exists some absolute constants $C > 0, c > 0$ such that for any $\tau \in \mathbb{Z}^+$, and $0 < \delta < 1/e$, for any $\theta > 0$, with probability at least $1 - \delta$ given $\mathcal{F}_{t-1}$,

$$\|E_\tau\|_F \leq c \cdot \theta \sum_{t=0}^{\tau-1} dm \log^3 \left(\frac{C\tau m}{\delta} \right) \|V_t\|_F^2 + \frac{1}{\theta} \log \left(\frac{Cm^2 d \tau}{\delta} \right).$$

Proof. Let $V_t = P\Sigma Q^\top = \sum_{i=1}^r \sigma_i p_i q_i^\top$ be the singular value decomposition of V_t . Extend $\{q_1, \dots, q_r\}$ to an orthonormal basis $\{q_1, \dots, q_m\}$ of $\mathbb{R}^m$. Define the events

$$A_t = \left\{ \sqrt{d}L_\delta < \|a_t\| < \sqrt{3d \log(C/\delta)} \right\}$$

and

$$B_{t,i} = \left\{ L_\delta < |b_t^\top q_i| < \sqrt{3 \log(C/\delta)} \right\}, \quad B_t = \bigcap_{i=1}^m B_{t,i}.$$

Let R be an orthogonal matrix $RR^\top = R^\top R = I$, and notice that Ra has the same distribution as a , and that A_t is the same event when $\|a_t\|$ is replaced with $\|Ra_t\|$. Therefore,

$$R \mathbb{E}_{\mathcal{F}_{t-1}} \left[a_t a_t^\top; A_t \right] R^\top = \mathbb{E}_{\mathcal{F}_{t-1}} \left[(Ra_t)(Ra_t)^\top; A_t \right] = \mathbb{E}_{\mathcal{F}_{t-1}} \left[a_t a_t^\top; A_t \right]$$

Now, if M is a symmetric matrix such that $RM R^\top = M$ for all R , it must be that M is a multiple of the identity. Therefore,

$$\mathbb{E}_{\mathcal{F}_{t-1}} \left[a_t a_t^\top; A_t \right] = \alpha I_d.$$

We then get

$$\begin{aligned}\alpha d &= \text{Tr}\left(\mathbb{E}_{\mathcal{F}_{t-1}}\left[a_t a_t^\top; A_t\right]\right) = \mathbb{E}_{\mathcal{F}_{t-1}}\left[\|a_t\|^2; A_t\right] \\ &= \mathbb{E}_{\mathcal{F}_{t-1}}\left[(\|a_t\|^2 - d); A_t\right] + d\mathbb{P}(A_t) = d\mathbb{P}(A_t)\end{aligned}$$

The most important fact for this entire proof, is to notice that the events $B_{t,i}$ are independent.

This is because, since $q_i \perp q_j$ for $i \neq j$, the $b_t^\top q_i$ and $b_t^\top q_j$ are independent given $\mathcal{F}_{t-1}$.

Thus, if $i \neq j$,

$$\begin{aligned}\mathbb{E}_{\mathcal{F}_{t-1}}\left[(Q^\top b_t)(Q^\top b_t)_{i,j}^\top; B_t\right] &= \mathbb{E}_{\mathcal{F}_{t-1}}\left[(q_i^\top b_t)(q_j^\top b_t); B_t\right] \\ &= \mathbb{P}_{\mathcal{F}_{t-1}}\left(\bigcap_{k \neq i,j} B_{t,k}\right) \cdot \mathbb{E}_{\mathcal{F}_{t-1}}\left[q_i^\top b_t; B_{t,i}\right] \cdot \mathbb{E}_{\mathcal{F}_{t-1}}\left[q_j^\top b_t; B_{t,j}\right] \\ &= 0\end{aligned}$$

because if Z is $\mathcal{N}(0, 1)$ and $A \in \sigma(Z)$ is a symmetric event, we have $\mathbb{E}[Z; A] = 0$. When $i = j$,

$$\mathbb{E}_{\mathcal{F}_{t-1}}\left[(q_i^\top b_t)^2; B_t\right] = \mathbb{P}_{\mathcal{F}_{t-1}}\left(\bigcap_{k \neq i} B_{t,k}\right) \cdot \mathbb{E}_{\mathcal{F}_{t-1}}\left[(q_i^\top b_t)^2; B_{t,i}\right] = \mathbb{P}_{\mathcal{F}_{t-1}}(B_t)$$

by our judicious choice of $B_{t,i}$ (cf. the lemma before this).

As $Q \in \mathcal{F}_{t-1}$, we thus have

$$\begin{aligned}\mathbb{E}_{\mathcal{F}_{t-1}}\left[Q^\top b_t b_t^\top; B_t\right] &= \mathbb{E}_{\mathcal{F}_{t-1}}\left[(Q^\top b_t)(Q^\top b_t)^\top; B_t\right] \cdot Q^\top \\ &= \mathbb{P}_{\mathcal{F}_{t-1}}(B)I \cdot Q^\top = \mathbb{P}_{\mathcal{F}_{t-1}}(B)Q^\top.\end{aligned}$$

Therefore,

$$\mathbb{E}_{\mathcal{F}_{t-1}}\left[V_t - a_t a_t^\top V_t b_t b_t^\top; A_t, B_t\right] = V_t \mathbb{P}_{\mathcal{F}_{t-1}}(A_t \cap B_t) - \mathbb{E}_{\mathcal{F}_{t-1}}\left[a_t a_t^\top P \Sigma Q^\top b_t b_t^\top; A_t, B_t\right]$$

Now,

$$\begin{aligned}\mathbb{E}_{\mathcal{F}_{t-1}}[a_t a_t^\top P \Sigma Q^\top b_t b_t^\top; A_t, B_t] &= \mathbb{E}_{\mathcal{F}_{t-1}}[a_t a_t^\top; A_t] P \Sigma \cdot \mathbb{E}_{\mathcal{F}_{t-1}}[Q^\top b_t b_t^\top; B_t] \\ &= P \Sigma Q^\top = V_t\end{aligned}$$

Therefore, upon defining

$$Q_t = (V_t - a_t a_t^\top V_t b_t b_t^\top) \mathbb{1}_{A_t \cap B_t}$$

we have

$$\mathbb{E}_{\mathcal{F}_{t-1}}[Q_t] = 0.$$

The next part of the proof is showing that Q_t is $\text{nSG}(C\sqrt{mn}\text{polylog}(dm/\delta) \|V\|_F)$.

First, notice that on B_t ,

$$\|b_t\|^2 = \sum_{i=1}^m |q_i^\top b_t| \leq 3m \log(C/\delta)$$

and recall that A_t was defined so that

$$\|a_t\| \leq \sqrt{3d \log(C/\delta)}.$$

So on $A_t \cap B_t$ we have

$$\left\| a_t a_t^\top V_t b_t b_t^\top \right\|_F = |a_t^\top V b_t| \|a_t\| \|b_t\| \leq |a_t^\top V b_t| \sqrt{9md \log^2(C/\delta)}$$

Our main goal now is to show that $a_t^\top V b_t$ is subgaussian with parameter $\tilde{O}(\|V\|_F)$. To simplify notation, we will drop the ts and conditioning, it won't be important for the argument. Recall that for $Z \sim \mathcal{N}(0, \sigma^2)$, if $t\sigma^2 < \frac{1}{2}$,

$$\mathbb{E}[e^{tZ^2}] = (1 - 2t\sigma^2)^{-1/2}$$

Then given b , since $a^\top V b$ has distribution $\mathcal{N}(0, \|Vb\|^2)$, we see immediately that whenever

$$t \|Vb\|^2 < \frac{1}{4},$$

$$\mathbb{E}\left[e^{t(a^\top Vb)^2} \mid b\right] = (1 - 2t \|Vb\|^2)^{-1/2} \leq e^{2t\|Vb\|^2}$$

where the latter bound comes from, if $x \leq \frac{1}{4}$,

$$\log\left((1 - 2x)^{-1/2}\right) = -\frac{1}{2} \log(1 - 2x) \leq \frac{1}{2} \cdot \frac{2x}{1 - 2x} \leq 2x$$

While it is now tempting to use $\|Vb\| \leq \|V\|_F \cdot \|b\|$, this leads to a bad bound. Instead, observe that Vb is $\mathcal{N}(0, VV^\top)$ distributed.

Now recall that for $Z \sim \mathcal{N}(0, A)$, if $t < 1/(2\lambda_{\max}(A))$,

$$\mathbb{E}\left[e^{t\|Z\|^2}\right] = \det(I - 2tA)^{-1/2}$$

Therefore, as long as $t < 1/(8\sigma_{\max}(V))$, we have that

$$\begin{aligned} \mathbb{E}\left[e^{2t\|Vb\|^2}\right] &= \prod_{i=1}^r (1 - 4t\sigma_i^2)^{-1/2} = \exp\left(\sum_{i=1}^r -\frac{1}{2} \log(1 - 4t\sigma_i^2)\right) \\ &\leq \exp\left(4t \sum_{i=1}^r \sigma_i^2\right) = \exp\left(4t \|V\|_F^2\right) \end{aligned}$$

We are now almost done. Observe, on the event B ,

$$\|Vb\|^2 = b^\top V^\top Vb = \sum_{i=1}^r \sigma_i^2 b^\top q_i \leq 3 \log(C/\delta) \sum_{i=1}^r \sigma_i^2 = 3 \log(C/\delta) \|V\|_F^2$$

Now, $t \|Vb\|^2 \leq t 3 \|V\|_F^2 \log(C/\delta)$, so choosing $t = 1/(24 \|V\|_F^2 \log(C/\delta))$, we will satisfy all the bounds above ($\sigma_{\max}(V) \leq \|V\|_F$), and get

$$\mathbb{E}\left[e^{t(a^\top Vb)^2}\right] = \mathbb{E}\left[\mathbb{E}\left[e^{t(a^\top Vb)^2} \mid b\right]\right] \leq e^{4t\|V\|_F^2} \leq \exp\left(\frac{1}{6 \log(C/\delta)}\right) \leq e^{1/6} \leq 2$$

Therefore, $a_t^\top V_t b_t$ is $\text{nSG}(\sqrt{24 \log(C/\delta)} \|V\|_F)$. Combining the previous bounds on $\|a\|$ and $\|b\|$, we get

$$\mathbb{E}_{\mathcal{F}_{t-1}} \left[\exp\left(\frac{\|a_t a_t^\top V_t b_t b_t^\top \mathbb{1}_{A_t \cap B_t}\|_F^2}{216md \log^3(C/\delta) \|V\|_F^2}\right) \right] \leq 2$$

Therefore, using $\|Q_t\|_F^2 \leq \|V_t\|^2 + \|a_t a_t^\top V_t b_t b_t^\top \mathbb{1}_{A_t \cap B_t}\|$, for some absolute constant c' ,

$$\mathbb{E}_{\mathcal{F}_{t-1}} \left[\exp \left(\frac{\|Q_t\|_F^2}{c' m d \log^3(C/\delta) \|V\|_F^2} \right) \right] \leq 2$$

This proves that Q_t is nSG($c' \log^{3/2}(C/\delta) \|V\|_F$).

Thus, by Lemma 1, for some absolute constant c , with probability at least $1 - \delta$,

$$\left\| \sum_{t=0}^{\tau-1} Q_t \right\| \leq c \cdot \theta \sum_{t=0}^{\tau-1} m d \log^3(C/\delta) \|V_t\|_F^2 + \frac{1}{\theta} \log \frac{2mn}{\delta}.$$

Now, on $\bigcap_{t=0}^{\tau-1} A_t \cap B_t$,

$$\sum_{t=0}^{\tau-1} Q_t = \sum_{t=0}^{\tau-1} (V_t - a_t a_t^\top V_t b_t b_t^\top)$$

As $\mathbb{P}_{\mathcal{F}_{t-1}}(B_{t,i}) \geq 1 - \delta$, by union bound $\mathbb{P}_{\mathcal{F}_{t-1}}(B_t) \geq 1 - m\delta$. Further, recall that $\mathbb{P}_{\mathcal{F}_{t-1}}(A_t) \geq 1 - \delta$. Therefore,

$$\mathbb{P}_{\mathcal{F}_{t-1}} \left(\bigcap_{t=0}^{\tau-1} A_t \cap B_t \right) \geq 1 - \tau(m+1)\delta$$

so, with probability at least $1 - \delta - \tau(m+1)\delta$,

$$\|E_\tau\| \leq c \cdot \theta \sum_{t=0}^{\tau-1} m d \log^3(C/\delta) \|V_t\|_F^2 + \frac{1}{\theta} \log \left(\frac{2md}{\delta} \right).$$

After rescaling δ to $\delta/(1 + \tau(m+1))$, we have arrived at the desired result. $\square$

We will need this definition to bound the quadratic error.

Definition 5.9. Random variable $X \in \mathbb{R}$ is SW(K, α) for some $K, \alpha > 0$ if

$$\mathbb{P}(|X| \geq s) \leq 2 \exp \left(-(s/K)^{1/\alpha} \right) \quad \forall s \geq 0$$

We will need the following lemmas.

Lemma 5.10. Suppose X and Y are SW(K_X, α) and SW(K_Y, α) respectively. Then XY is SW($C(K_X \cdot K_Y), 2\alpha$) and $X + Y$ is SW($C(K_X + K_Y), \alpha$) for some absolute constant C .

Lemma 5.11. *Suppose $X_1, \dots, X_n$ are identically distributed $SW(K', \alpha)$. Then for some absolute constant $c > 0$, for all $s \geq ncK'$, we have*

$$\mathbb{P}\left(\left|\sum_{i=1}^n X_i\right| \geq s\right) \leq \exp\left(-\left(\frac{s}{ncK'}\right)^{1/\alpha}\right).$$

We will need to frequently bound sums of $\|P_t\|^{2k}$.

Lemma 5.12. *Suppose $a_0, \dots, a_{\tau-1} \sim \mathcal{N}(0, I_d)$, and $b_0, \dots, b_{\tau-1} \sim \mathcal{N}(0, I_m)$. Then for any $k > 0$ there exists absolute constants $c, C > 0$ such that for any $0 < \delta < 1$, w.p. $\geq 1 - \delta$,*

$$\sum_{t=0}^{\tau-1} \left\|a_i b_i^\top\right\|_F^{2k} \leq \tau C c^{2k} d^k m^k (1 + \log(1/\delta)^{2k})$$

When $\delta < 1/e$,

$$\sum_{t=0}^{\tau-1} \left\|a_i b_i^\top\right\|_F^{2k} \leq 2\tau C c^{2k} d^k m^k \log(1/\delta)^{2k}.$$

Proof. $(a_i)_k^2$ and $(b_j)_l^2$ are $SW(1, 1)$. Thus their product is $SW(c, 2)$. Thus $\|a_i b_i^\top\|_F^2 = \sum_{k,l} (a_i)_k^2 (b_i)_l^2$ is $SW(cmd, 2)$ for some absolute constant c . Thus, $\|a_i b_i^\top\|_F^{2k}$ is $SW(c^k m^k d^k, 2k)$. Therefore by the lemma, for $s \geq \tau C c^k m^k d^k$,

$$\mathbb{P}\left(\sum_{t=0}^{\tau-1} \left\|a_i b_i^\top\right\|_F^{2k} \geq s\right) \leq \exp\left(-\left(\frac{s}{\tau C c^k d^k m^k}\right)^{1/2k}\right)$$

the lemma follows by choosing $s = (1 + \log(1/\delta)^{2k}) \tau C c^k d^k m^k$. $\square$

5.3 Function Decrease

First, we prove the following crucial lemma.

Lemma 5.13 (Function decrease for rank r symmetric EGGROLL). *Suppose at each time t , the algorithm performs the update step with rank $r \leq dm$,*

$$X_{t+1} = X_t - \eta g_u^r(X_t)$$

where

$$g_\sigma^r(X_t) := \frac{1}{r} \sum_{i=1}^r \frac{f(X_t + \sigma P_{t,i}) - f(X_t - \sigma P_{t,i})}{2\sigma} P_{t,i}$$

where $P_{t,i} = a_{t,i} b_{t,i}^\top$ where each $a_{t,i}, b_{t,i}$ are drawn i.i.d. from $\mathcal{N}(0, I_d)$, and $\mathcal{N}(0, I_m)$ respectively, and $\sigma > 0$ is the smoothing radius.

Then there exists absolute constants $c > 0, C \geq 1$ such that, for any $T \in \mathbb{Z}^+$ and $1 \leq \tau' \leq T$, and $0 < \delta < 1/e$, then upon defining $\mathcal{H}_\tau(\delta)$ to be the event on which the inequality

$$\begin{aligned} f(X_\tau) - f(X_0) \leq & -\frac{\eta}{2} \sum_{t=0}^{\tau-1} \frac{1}{r} \sum_{i=1}^r |\langle \nabla f(X_t), P_{t,i} \rangle|^2 + \tau C c^4 \eta \sigma^2 L^2 (dm)^2 \log\left(\frac{T}{\delta}\right)^4 \\ & + \eta^2 \frac{cdm\chi^4}{r} \sum_{t=0}^{\tau-1} \|\nabla f(X_t)\|^2 + \eta^2 \tau C c^6 \sigma^2 L^3 (dm)^3 \log\left(\frac{T}{\delta}\right)^6 \end{aligned}$$

holds, (where $\chi := \log(Cm^2 dr T / \delta)$), we have

$$\mathbb{P}\left(\bigcap_{\tau=1}^{\tau'} \mathcal{H}_\tau(\delta)\right) \geq 1 - \frac{3\tau'}{T} \delta$$

Proof. For each $t \in \{-1, \dots, \tau\}$, let

$$\mathcal{F}_t = \sigma(X_0, \{a_{0,i}\}_{i=1}^r, \dots, \{a_{t,i}\}_{i=1}^r, \{b_{0,i}\}_{i=1}^r, \dots, \{b_{t,i}\}_{i=1}^r)$$

be the sigma-algebra containing the information available at time t , and as usual write

$$P_{t,i} = a_{t,i} b_{t,i}^\top.$$

Observe that, by mean value theorem, there exists some $s_{t,i} \in [-1, 1]$ such that

$$g_\sigma(X_t) = \frac{1}{r} \sum_{i=1}^r \frac{f(X_t + \sigma P_{t,i}) - f(X_t - \sigma P_{t,i})}{2\sigma} P_{t,i} = \frac{1}{r} \sum_{i=1}^r \nabla f(X_t + \sigma s_{t,i} P_{t,i}) P_{t,i}$$

Then we have that

$$X_{t+1} - X_t = -\eta \cdot \frac{1}{r} \sum_{i=1}^r (\langle \nabla f(X_t), P_{t,i} \rangle P_{t,i} + \langle \nabla f(X_t + \sigma s_{t,i} P_{t,i}) - \nabla f(X_t), P_{t,i} \rangle P_{t,i})$$

$$\begin{aligned}
& f(X_{t+1}) - f(X_t) \\
& \leq \langle X_{t+1} - X_t, \nabla f(X_t) \rangle + \frac{L}{2} \|X_{t+1} - X_t\|^2 \\
& \leq -\eta \cdot \frac{1}{r} \sum_{i=1}^r |\langle \nabla f(X_t), P_{t,i} \rangle|^2 \\
& \quad - \eta \cdot \frac{1}{r} \sum_{i=1}^r \langle \nabla f(X_t), P_{t,i} \rangle \langle \nabla f(X_t + \sigma_{t,i} P_{t,i}) - \nabla f(X_t), P_{t,i} \rangle \\
& \quad + \frac{L\eta^2}{2} \left\| \frac{1}{r} \sum_{i=1}^r (\langle \nabla f(X_t), P_{t,i} \rangle P_{t,i} + \langle \nabla f(X_t + \sigma_{t,i} P_{t,i}) - \nabla f(X_t), P_{t,i} \rangle P_{t,i}) \right\|^2
\end{aligned}$$

Now notice, by using $ab \leq (a^2 + b^2)/2$,

$$\begin{aligned}
& -\frac{\eta}{r} \sum_{i=1}^r \langle \nabla f(X_t), P_{t,i} \rangle \langle \nabla f(X_t + \sigma_{t,i} P_{t,i}) - \nabla f(X_t), P_{t,i} \rangle \\
& \leq \frac{\eta}{2r} \sum_{i=1}^r \left(|\langle \nabla f(X_t), P_{t,i} \rangle|^2 + \sigma^2 L^2 \|P_{t,i}\|^4 \right)
\end{aligned}$$

Also, by $\|a + b\|^2 \leq 2\|a\|^2 + 2\|b\|^2$,

$$\begin{aligned}
& \left\| \frac{1}{r} \sum_{i=1}^r (\langle \nabla f(X_t), P_{t,i} \rangle P_{t,i} + \langle \nabla f(X_t + \sigma_{t,i} P_{t,i}) - \nabla f(X_t), P_{t,i} \rangle P_{t,i}) \right\|^2 \\
& \leq 2 \left\| \frac{1}{r} \sum_{i=1}^r \langle \nabla f(X_t), P_{t,i} \rangle P_{t,i} \right\|^2 + \frac{2}{r} \sum_{i=1}^r \|\langle \nabla f(X_t + \sigma_{t,i} P_{t,i}) - \nabla f(X_t), P_{t,i} \rangle P_{t,i}\|^2 \\
& \leq 2 \left\| \frac{1}{r} \sum_{i=1}^r \langle \nabla f(X_t), P_{t,i} \rangle P_{t,i} \right\|^2 + \frac{2\sigma^2 L^2}{r} \sum_{i=1}^r \|P_{t,i}\|^6
\end{aligned}$$

Therefore,

$$\begin{aligned}
f(X_{t+1}) - f(X_t) & \leq -\frac{\eta}{2r} \sum_{i=1}^r |\langle \nabla f(X_t), P_{t,i} \rangle|^2 + \frac{\eta\sigma^2 L^2}{2r} \sum_{i=1}^r \|P_{t,i}\|^4 \\
& \quad + L\eta^2 \left\| \frac{1}{r} \sum_{i=1}^r \langle \nabla f(X_t), P_{t,i} \rangle P_{t,i} \right\|^2 + \frac{\eta^2 L^3 \sigma^2}{r} \sum_{i=1}^r \|P_{t,i}\|^6
\end{aligned}$$

Now,

$$\left\| \frac{1}{r} \sum_{i=1}^r \langle \nabla f(X_t), P_{t,i} \rangle P_{t,i} \right\|^2 \leq 2 \left\| \frac{1}{r} \sum_{i=1}^r (\nabla f(X_t) - \langle \nabla f(X_t), P_{t,i} \rangle P_{t,i}) \right\|^2 + 2 \|\nabla f(X_t)\|^2$$

By our proposition, with probability $1 - \delta$,

$$\left\| \sum_{i=1}^r (\nabla f(X_t) - \langle \nabla f(X_t), P_{t,i} \rangle P_{t,i}) \right\| \leq c\theta dmr \log^3 \left(\frac{Crm}{\delta} \right) \|\nabla f(X_t)\|^2 + \frac{1}{\theta} \log \left(\frac{Cm^2 dr}{\delta} \right)$$

As $\nabla f(X_t)$ is fixed given $\mathcal{F}_{t-1}$, optimizing over θ , this is at most

$$\sqrt{cdmr} \log(Cm^2 dr/\delta)^2 \|\nabla f(X_t)\|.$$

Therefore for some irrelevant absolute constant c ,

$$\left\| \frac{1}{r} \sum_{i=1}^r \langle \nabla f(X_t), P_{t,i} \rangle P_{t,i} \right\|^2 \leq c \frac{dm}{r} \log(Cm^2 dr/\delta)^4 \|\nabla f(X_t)\|^2 \quad (5.1)$$

By union bound, this will hold jointly for all $0 \leq t \leq \tau' - 1$ with probability at least $1 - \tau'\delta$.

Now by old lemma, for some absolute constants C, c possibly different from before, with probability at least $1 - \delta$,

$$\frac{1}{r} \sum_{t=0}^{\tau-1} \sum_{i=1}^r \|P_{t,i}\|^{2k} \leq 2\tau C c^{2k} (dm)^k \log(1/\delta)^{2k}.$$

If k is fixed, by union bound, this will hold jointly for all $1 \leq \tau \leq \tau'$ with probability $1 - \tau'\delta$.

Therefore, it will hold for both $k = 2$ and 3 , for all $1 \leq \tau \leq \tau'$, with probability at least $1 - 2\tau'\delta$.

Therefore,

$$\begin{aligned} f(X_\tau) - f(X_0) &\leq -\frac{\eta}{2r} \sum_{t=0}^{\tau-1} \sum_{i=1}^r |\langle \nabla f(X_t), P_{t,i} \rangle|^2 + \tau C c^4 \eta \sigma^2 L^2 (dm)^2 \log(1/\delta)^4 \\ &\quad + \eta^2 \frac{cdm}{r} \log(Cm^2 dr/\delta)^4 \sum_{t=0}^{\tau-1} \|\nabla f(X_t)\|^2 + \eta^2 \tau C c^6 \sigma^2 L^3 (dm)^3 \log(1/\delta)^6 \end{aligned}$$

holds for all $1 \leq \tau \leq \tau'$ with probability at least $1 - 3\tau\delta$. Equivalently,

$$\mathbb{P}\left(\bigcap_{\tau=1}^{\tau'} \mathcal{H}_\tau(\delta)\right) \geq 1 - 3\tau'\delta$$

Similarly, we see that, as when τ is fixed we need only Equation ?? to hold jointly over $0 \leq t \leq \tau - 1$, while the bound on the perturbation norms hold with probability at least $1 - 2\delta$, so,

$$\mathbb{P}(\mathcal{H}_\tau(\delta)) \geq 1 - (\tau + 2)\delta$$

After rescaling δ to δ/T , we have arrived at the desired result. $\square$

5.4 Gradients Don't Move too Quickly

We will need the following Bernstein inequality for sub-exponential random variables (Theorem 2.8.1).

Lemma 5.14 (Bernstein concentration inequality). *Let $X_1, \dots, X_n$ be independent, mean-zero, σ -subexponential random variables, i.e.*

$$\mathbb{E}\left[\exp\left(\frac{|X_i|}{\sigma}\right)\right] \leq 2 \quad \text{for all } i \in [n].$$

Then there exists an absolute constant c such that for any $s \geq 0$,

$$\mathbb{P}\left(\sum_{i=1}^n X_i \geq s\right) \leq \exp\left(-c \min\left\{\frac{s^2}{n\sigma^2}, \frac{s}{\sigma}\right\}\right).$$

We then prove the following lemma.

Lemma 5.15. *There exists an absolute constant $c_2 \geq 1$ such that, if*

$$t_f(\delta) = \frac{c_2}{m} \log\left(\frac{T}{\delta}\right), \quad \delta > 0,$$

and

$$\mathcal{B}_{t_0}(\delta; k) = \bigcup_{t=t_0}^{t_0+k-1} \left\{ \frac{1}{m} \sum_{i=1}^m |a_{t,i}^\top \nabla f(X_t) b_{t,i}|^2 \geq \frac{1}{2} \|\nabla f(X_t)\|^2 \right\}$$

we have

$$\mathbb{P}(\mathcal{B}_{t_0}(\delta; k)) \geq 1 - \frac{\delta}{T}$$

for $0 < \delta < 1$, $t_0 \geq 0$ and $k \geq t_f(\delta)$.

Proof. By using the same technique from the proof of Theorem 3, we know that the set of random variables $\left\{ \|\nabla f(X_t)\|^2 - |a_{t,i}^\top \nabla f(X_t) b_{t,i}|^2 \right\}_{i=1}^m$ are independent, $c \|\nabla f(X_t)\|^2$ subexponential for some absolute constant $c > 0$ given $\mathcal{F}_{t-1}$. Now define

$$E_t = \left\{ \frac{1}{m} \sum_{i=1}^m |a_{t,i}^\top \nabla f(X_t) b_{t,i}|^2 < \frac{1}{2} \|\nabla f(X_t)\|^2 \right\}$$

Therefore,

$$\begin{aligned} \mathbb{P}_{\mathcal{F}_{t-1}}(E_t) &= \mathbb{P}_{\mathcal{F}_{t-1}} \left(\frac{1}{m} \sum_{i=1}^m |a_{t,i}^\top \nabla f(X_t) b_{t,i}|^2 \geq \|\nabla f(X_t)\|^2 \right) \\ &= \mathbb{P}_{\mathcal{F}_{t-1}} \left(\sum_{i=1}^m \left(\|\nabla f(X_t)\|^2 - |a_{t,i}^\top \nabla f(X_t) b_{t,i}|^2 \right) > \frac{m}{2} \|\nabla f(X_t)\|^2 \right) \\ &\leq \exp \left(-c' \min \left\{ \frac{m}{4c^2}, \frac{m}{2c} \right\} \right) \leq \exp(-c''m) \end{aligned}$$

for some absolute constant c'' , where the last line follows by using Bernstein concentration inequality.

Now,

$$\begin{aligned} \mathbb{P} \left(\bigcap_{t=t_0}^{t_0+k-1} E_t \right) &= \mathbb{P} \left(\bigcap_{t=t_0}^{t_0+k-1} \left\{ \frac{1}{m} \sum_{i=1}^m |a_{t,i}^\top \nabla f(X_t) b_{t,i}|^2 < \frac{1}{2} \|\nabla f(X_t)\|^2 \right\} \right) \\ &= \mathbb{E} \left[\prod_{t=t_0}^{t_0+k-2} \mathbb{1}_{E_t} \mathbb{E}_{\mathcal{F}_{t_0+k-2}} [\mathbb{1}_{E_{t_0+k-1}}] \right] \leq \cdots \leq \exp(c''mk) \end{aligned}$$

therefore by choosing $k \geq t_f(\delta)$ we get

$$\mathbb{P}\left(\bigcap_{t=t_0}^{t_0+k-1} \left\{ \frac{1}{m} \sum_{i=1}^m |a_{t,i} \nabla f(X_t) b_{t,i}|^2 < \frac{1}{2} \|\nabla f(X_t)\|^2 \right\}\right) \leq \frac{\delta}{T},$$

which completes the proof. $\square$

Lemma 5.16. *Define*

$$\mathcal{A}_t(\delta) := \left\{ \|\nabla f(X_{t+1}) - \nabla f(X_t)\| \leq \frac{\|\nabla f(X_t)\|}{8t_f(\delta)} + \eta\sigma L^2 \cdot \frac{1}{r} \sum_{i=1}^r \|P_{t,i}\|^3 \right\}$$

where $t_f(\delta)$ is defined above. Then there exists an absolute constant c_2 such that, whenever η satisfies,

$$\eta L c_2 \sqrt{\frac{dm}{r}} \log\left(\frac{CdmrT}{\delta}\right)^2 \leq \frac{1}{8t_f(\delta)}, \quad (5.2)$$

we have

$$\mathbb{P}(\mathcal{A}_t(\delta)) \geq 1 - \frac{\delta}{T}$$

for any $0 < \delta \leq 1/e$ and $T \in \mathbb{Z}^+$.

Proof. By L -lipschitz of ∇f ,

$$\begin{aligned} \|\nabla f(X_{t+1}) - \nabla f(X_t)\| &\leq L \|X_{t+1} - X_t\| \\ &\leq \eta L \left\| \frac{1}{r} \sum_{i=1}^r a_{t,i} a_{t,i}^\top \nabla f(X_t) b_{t,i} b_{t,i}^\top \right\| \\ &\quad + \eta L \left\| \frac{1}{r} \sum_{i=1}^r \langle \nabla f(X_t + \sigma s_{t,i} P_{t,i}) - \nabla f(X_t), P_{t,i} \rangle P_{t,i} \right\| \end{aligned}$$

By L -lipschitz of $\nabla f(X_t)$ and the Cauchy-Schwarz inequality,

$$\left\| \frac{1}{r} \sum_{i=1}^r \langle \nabla f(X_t + \sigma s_{t,i} P_{t,i}) - \nabla f(X_t), P_{t,i} \rangle P_{t,i} \right\| \leq L\sigma \cdot \frac{1}{r} \sum_{i=1}^r \|P_{t,i}\|^3$$

Now for some absolute constant $c_2 > 0$ by Theorem ??, with probability at least $1 - \delta/T$,

$$\left\| \frac{1}{r} \sum_{i=1}^r a_{t,i} a_{t,i}^\top \nabla f(X_t) b_{t,i} b_{t,i}^\top \right\| \leq c_2 \log(CdmrT/\delta)^2 \sqrt{\frac{dm}{r}} \|\nabla f(X_t)\|$$

plugging in our condition on η , we arrive at the statement of the lemma. $\square$

5.5 Progress Lower Bound

This next very important lemma is the key that will let us lower bound

$$\sum_{t=0}^{\tau-1} \frac{1}{r} \sum_{i=1}^r |\langle \nabla f(X_t), P_t \rangle|^2 \geq \tilde{\Omega} \left(\sum_{t=0}^{\tau-1} \|\nabla f(X_t)\| \right)$$

in some sense.

Denote

$$\mathcal{E}(t_1, t_2, \delta) = \bigcap_{t=t_1}^{t_1+t_2-1} \left\{ \|\nabla f(X_t)\| \geq 8t_f(\delta)\eta\sigma L^2 \cdot \frac{1}{r} \sum_{i=1}^r \|P_{t,i}\|^3 \right\}$$

Lemma 5.17. *Let $0 < \delta \leq 1/e$ and $T \in \mathbb{Z}^+$ satisfying $T \geq 2t_f(\delta) + 1$. Consider any positive integer $t'_f \leq 2t_f(\delta)$, and any $t_0 \in \{0, \dots, T - 1 - t'_f\}$. Suppose η satisfies the condition above, then on*

$$\mathcal{E}(t_0, t'_f, \delta) \cap \left(\bigcap_{t=t_0}^{t_0+t'_f-1} \mathcal{A}_t(\delta) \right)$$

we have

$$\frac{1}{2} \|\nabla f(X_0)\| \leq \|\nabla f(X_t)\| \leq 2 \|\nabla f(X_t)\|$$

for all $t \in \{t_0, \dots, t_0 + t'_f - 1\}$.

Proof. By plugging in the definition for $\mathcal{E}_t(\delta)$ into the definition of $\mathcal{A}_t(\delta)$, we see that on the prescribed event,

$$\|\nabla f(X_{t+1}) - \nabla f(X_t)\| \leq \frac{\|\nabla f(X_t)\|}{4t_f(\delta)}$$

therefore

$$\left(1 - \frac{1}{4t_f(\delta)}\right) \|\nabla f(X_t)\| \leq \|\nabla f(X_{t+1})\| \leq \left(1 + \frac{1}{4t_f(\delta)}\right) \|\nabla f(X_t)\|$$

which leads to

$$\left(1 - \frac{1}{4t_f(\delta)}\right)^{t-t_0} \|\nabla f(X_0)\| \leq \|\nabla f(X_t)\| \leq \left(1 + \frac{1}{4t_f(\delta)}\right)^{t-t_0} \|\nabla f(X_0)\|.$$

for all $t \in \{t_0, \dots, t_0 + t'_f - 1\}$. Now using $\log(1+x) \leq x$ for $x \geq 0$ and $-\log(1-x) \leq x/(1-x)$ for $0 \leq x < 1$, we know that

$$\left(1 - \frac{1}{4x}\right)^{2x} \geq e^{-2/3} \geq \frac{1}{2}, \quad \left(1 + \frac{1}{4x}\right)^{2x} \leq e^{1/2} \leq 2$$

Now recalling that $t'_f \leq 2t_f(\delta)$, we are done. $\square$

Lemma 5.18. *There exists an absolute constant $c > 0$ such that for any $t \in \mathbb{N}$, the event*

$$\mathcal{G}_t(\delta) := \left\{ \frac{1}{r} \sum_{i=1}^r \|P_{t,i}\|^3 \leq c(dm)^{3/2} \log(T/\delta)^3 \right\}$$

holds with probability at least $1 - \delta/T$ for $0 < \delta \leq 1/e$.

Proof. Obvious from Lemma ?? and the Cauchy-Schwarz inequality. $\square$

Lemma 5.19. *Let $0 < \delta \leq 1/e$ and $T \in \mathbb{Z}^+$ such that $T > 2t_f(\delta) + 1$. Consider any positive integer $t'_f \leq 2t_f(\delta)$ and $t_0 \in \{0, \dots, T - 1 - t'_f\}$. Suppose η satisfies the condition above. Then on the event*

$$\mathcal{E}^c(t_0, t'_f, \delta) \cap \left(\bigcap_{t=t_0}^{t_0+t'_f-1} \mathcal{A}_t(\delta) \right) \cap \left(\bigcap_{t=t_0}^{t_0+t'_f-1} \mathcal{G}_t(\delta) \right)$$

we have

$$\|\nabla f(X_t)\| \leq 16t_f(\delta)\eta\sigma L^2 c(dm)^{3/2} \log(T/\delta)^3$$

for all $t \in \{t_0, \dots, t_0 + t'_f - 1\}$.

Proof. Let t' be the first time in $\{t_0, \dots, t_0 + t_f - 1\}$ such that the gradient is dominated, meaning

$$\|\nabla f(X_{t'})\| \leq 8t_f(\delta)\eta L^2\sigma \frac{1}{r} \sum_{i=1}^r \|P_{t,i}\|^3.$$

By definition of $\mathcal{G}_{t'}(\delta)$, we must have

$$\|\nabla f(X_{t'})\| \leq 8t_f(\delta)\eta L^2\sigma c(dm)^{3/2} \log(T/\delta)^3$$

For $j < t'$, using the idea from a previous lemma,

$$\|\nabla f(X_j)\| \leq 2\|\nabla f(X_{t'})\| \leq 16t_f(\delta)\eta L^2\sigma c(dm)^{3/2} \log(T/\delta)^3$$

On the other hand, for $t \in \{t', \dots, t_0 + t'_f - 1\}$, by using definition of $\mathcal{A}_t(\delta)$ and $\mathcal{G}_t(\delta)$, we get

$$\begin{aligned} \|\nabla f(X_{t+1})\| &\leq \left(1 + \frac{1}{8t_f(\delta)}\right) \|\nabla f(X_t)\| + \eta\sigma L^2 c(dm)^{3/2} \log(T/\delta)^3 \\ &= \left(1 - \frac{1}{8t_f(\delta)}\right)^{t+1-t'} \|\nabla f(X_{t'})\| \\ &\quad + \sum_{i=0}^{t-t'} \left(1 - \frac{1}{8t_f(\delta)}\right)^{t-t'-i} \eta\sigma L^2 c(dm)^{3/2} \log(T/\delta)^3 \\ &\leq \left(1 + \frac{1}{8t_f(\delta)}\right)^{t'_f} \|\nabla f(X_{t'})\| \\ &\quad + 8t_f(\delta) \left(\left(1 + \frac{1}{8t_f(\delta)}\right)^{t'_f} - 1 \right) \eta\sigma L^2 c(dm)^{3/2} \log(T/\delta)^3 \\ &\leq e^{1/4} 8t_f(\delta) \eta\sigma L^2 c(dm)^{3/2} \log(T/\delta)^3 \\ &\quad + 8t_f(\delta) (e^{1/4} - 1) \eta\sigma L^2 c(dm)^{3/2} \log(T/\delta)^3 \\ &\leq 16t_f(\delta) \eta\sigma L^2 c(dm)^{3/2} \log(T/\delta)^3 \end{aligned}$$

Where we used $t'_f \leq 2t_f(\delta)$ and $(1 + 1/(8x))^{2x} \leq e^{1/4}$ when $x > 0$. This completes the proof. $\square$

5.6 Putting it all Together

We are now almost ready to prove the main theorem. But first, we need the following lemma.

Lemma 5.20 (Function change for large τ). *Let $0 < \delta \leq 1/e$ and let $\tau \geq t_f(\delta)$ be arbitrary. Split $\{0, \dots, \tau - 1\}$ into $K := \tau/t_f(\delta)$ intervals $J_0, \dots, J_{K-1}$ of size $t_f(\delta)$. Let I_1 denote the set of indices where the gradient always dominates the noise term in J_k , i.e.*

$$I_1 := \left\{ k : \|\nabla f(X_t)\| \geq 8t_f(\delta)\eta\sigma L^2 \frac{1}{r} \sum_{i=1}^r \|P_{t,i}\|^3 \text{ for all } t \in J_k \right\}$$

Suppose we choose η satisfying

$$\eta \leq \frac{1}{8t_f(\delta)} \min \left\{ \sqrt{\frac{r}{dm}} \cdot \frac{1}{c_2 L \log(Cdmr\tau/\delta)^2}, \frac{r}{dm} \cdot \frac{1}{8c\chi^4} \right\}$$

Then on the event

$$\mathcal{E}_\tau(\delta) := \mathcal{H}_\tau(\delta) \cap \left(\bigcap_{t=0}^{\tau-1} \mathcal{A}_t(\delta) \right) \cap \left(\bigcap_{t=0}^{\tau-1} \mathcal{G}_t(\delta) \right) \cap \left(\bigcap_{t=0}^{\tau-1} \mathcal{B}_{kt_f(\delta)}(\delta; t_f(\delta)) \right)$$

we have the following upper bound for some absolute constants c, C ,

$$\begin{aligned} f(X_\tau) - f(X_0) &\leq - \sum_{k \in I_1} \frac{\eta}{4} \min_{t \in J_k} \|\nabla f(X_t)\|^2 + \tau \eta^4 \frac{cdm\chi^4}{r} t_f(\delta)^2 \sigma^2 L^4 (dm)^3 \log(T/\delta)^6 \\ &\quad + \tau C c^4 \eta \sigma^2 L^2 (dm)^2 \log\left(\frac{T}{\delta}\right)^4 + \eta^2 \tau C c^6 \sigma^2 L^3 (dm)^3 \log\left(\frac{T}{\delta}\right)^6. \end{aligned}$$

Furthermore,

$$\mathbb{P}(\mathcal{E}_\tau(\delta)) \geq 1 - \frac{(4\tau + 2)}{T} \delta.$$

Proof. Any interval $J_k = \{t_0, \dots, t_0 + t_f - 1\}$ belongs to one of the following two cases:

(Case 1) (Gradient dominates noise): This means for every $t \in J_k$,

$$\|\nabla f(X_t)\| \geq 8t_f(\delta)\eta\sigma L^2 \frac{1}{r} \sum_{i=1}^r \|P_{t,i}\|^3$$

By the Lemma ?? we have

$$\min_{t \in J_k} \|\nabla f(X_t)\| \geq \frac{1}{4} \max_{t \in J_k} \|\nabla f(X_t)\|$$

On the event $\mathcal{B}_{kt_f(\delta)}(\delta; t_f(\delta))$, there exists some $t \in J_k$ such that

$$\frac{1}{r} \sum_{i=1}^r |\langle P_{t,i}, \nabla f(X_t) \rangle|^2 \geq \frac{1}{2} \|\nabla f(X_t)\|^2$$

Therefore, using an average,

$$\begin{aligned} \frac{1}{4} \sum_{t \in J_k} \frac{1}{r} \sum_{i=1}^r |\langle P_{t,i}, \nabla f(X_t) \rangle|^2 &\geq \frac{1}{4} \min_{t \in J_k} \|\nabla f(X_t)\| \geq \frac{1}{64} \max_{t \in J_k} \|\nabla f(X_t)\|^2 \\ &\geq \frac{1}{64t_f(\delta)} \sum_{t \in J_k} \|\nabla f(X_t)\|^2 \end{aligned}$$

By choosing

$$\frac{\eta c d m \chi^4}{r} \leq \frac{1}{64t_f(\delta)}$$

we have

$$\begin{aligned} & -\frac{\eta}{2} \sum_{t \in J_k} \frac{1}{r} \sum_{i=1}^r |\langle P_{t,i}, \nabla f(X_t) \rangle|^2 + \frac{\eta^2 c d m \chi^4}{r} \sum_{t \in J_k} \|\nabla f(X_t)\|^2 \\ & \leq -\frac{\eta}{4} \sum_{t \in J_k} \frac{1}{r} \sum_{i=1}^r |\langle P_{t,i}, \nabla f(X_t) \rangle|^2 \leq -\frac{\eta}{4} \min_{t \in J_k} \|\nabla f(X_t)\|^2. \end{aligned}$$

(Case 2) (Gradient does not dominate noise): This means that there exists some $t \in J_k$ such that

$$\|\nabla f(X_t)\| \leq 8t_f(\delta)\eta\sigma L^2 \frac{1}{r} \sum_{i=1}^r \|P_{t,i}\|^3$$

Then by Lemma ??, we know that, for all $t \in J_k$,

$$\|\nabla f(X_t)\| \leq 16t_f(\delta)\eta\sigma L^2 c(dm)^{3/2} \log(T/\delta)^3$$

Every J_k falls into one of the above two cases. Now we can complete the proof.

$$\begin{aligned}
& -\frac{\eta}{2} \sum_{t=0}^{\tau-1} \frac{1}{r} \sum_{i=1}^r |\langle P_{t,i}, \nabla f(X_t) \rangle|^2 + \eta^2 \frac{cdm\chi^4}{r} \sum_{t=0}^{\tau-1} \|\nabla f(X_t)\|^2 \\
& = \sum_{k \in I_1} \left(-\frac{\eta}{2} \sum_{t \in J_k} \frac{1}{r} \sum_{i=1}^r |\langle P_{t,i}, \nabla f(X_t) \rangle|^2 + \eta^2 \frac{cdm\chi^4}{r} \sum_{t \in J_k} \|\nabla f(X_t)\|^2 \right) \\
& \quad + \sum_{k \in I_1^c} \left(-\frac{\eta}{2} \sum_{t \in J_k} \frac{1}{r} \sum_{i=1}^r |\langle P_{t,i}, \nabla f(X_t) \rangle|^2 + \eta^2 \frac{cdm\chi^4}{r} \sum_{t \in J_k} \|\nabla f(X_t)\|^2 \right) \\
& \leq - \sum_{k \in I_1} \frac{\eta}{4} \min_{t \in J_k} \|\nabla f(X_t)\|^2 + \sum_{k \in I_1^c} t_f(\delta) \left(\eta^4 \frac{cdm\chi^4}{r} 256 t_f(\delta)^2 \sigma^2 L^4 c(dm)^3 \log(T/\delta)^6 \right) \\
& \leq - \sum_{k \in I_1} \frac{\eta}{4} \min_{t \in J_k} \|\nabla f(X_t)\|^2 + \tau \eta^4 \frac{cdm\chi^4}{r} 256 t_f(\delta)^2 \sigma^2 L^4 c(dm)^3 \log(T/\delta)^6
\end{aligned}$$

and thus by the definition of $\mathcal{H}_\tau(\delta)$ in Lemma ??,

$$\begin{aligned}
f(X_\tau) - f(X_0) & \leq - \sum_{k \in I_1} \frac{\eta}{4} \min_{t \in J_k} \|\nabla f(X_t)\|^2 + \tau \eta^4 \frac{cdm\chi^4}{r} 256 t_f(\delta)^2 \sigma^2 L^4 c(dm)^3 \log(T/\delta)^6 \\
& \quad + \tau C c^4 \eta \sigma^2 L^2 (dm)^2 \log\left(\frac{T}{\delta}\right)^4 + \eta^2 \tau C c^6 \sigma^2 L^3 (dm)^3 \log\left(\frac{T}{\delta}\right)^6.
\end{aligned}$$

Furthermore,

$$\begin{aligned}
\mathbb{P}(\mathcal{E}_\tau(\delta)^c) & \leq \mathbb{P}(\mathcal{H}_\tau(\delta)^c) + \sum_{t=0}^{\tau-1} \mathbb{P}(\mathcal{A}_t(\delta)^c) + \sum_{t=0}^{\tau-1} \mathbb{P}(\mathcal{G}_t(\delta)^c) + \sum_{k=0}^{K-1} \mathbb{P}(\mathcal{B}_{kt_f(\delta)}(\delta; t_f(\delta))^c) \\
& \leq \frac{(\tau+2)}{T} \delta + \frac{\tau}{T} \delta + \frac{\tau}{T} \delta + \frac{K}{T} \delta \leq \frac{(4\tau+2)}{T} \delta
\end{aligned}$$

which completes the proof. $\square$

Theorem 5.21 (Symmetric EGGROLL Convergence). *Let $0 < \delta < 1/e$ be arbitrary.*

Suppose we choose σ, η and T such that

$$\begin{aligned}\sigma^2 &\leq \min \left\{ \frac{\varepsilon^2}{1024Cc^4L^2(dm)^2 \log(T/\delta)^4}, \frac{\varepsilon^2}{8096rL^3(dm)^2 \log(T/\delta)^2} \right\} \\ \eta &\leq \frac{1}{8t_f(\delta)} \min \left\{ \sqrt{\frac{r}{dm}} \cdot \frac{1}{cL \log(Cdmr\tau/\delta)^2}, \frac{r}{dm} \cdot \frac{1}{8c\chi^4} \right\}, \\ T &\geq \max \left\{ 4, \frac{512t_f(\delta)(f(X_0) - f^*)}{\eta\varepsilon^2} \right\}.\end{aligned}$$

Then with probability at least $1 - 5\delta$ there are at most $T/4$ iterations for which $\|\nabla f(X_t)\| \geq \varepsilon$.

Proof. Without loss of generality assume that $t_f(\delta)$ divides T and split $\{0, \dots, T-1\}$ into $K = T/t_f(\delta)$ intervals $J_0, \dots, J_{K-1}$. Let I_1 denote the set of indices k such that for every $t \in J_k$,

$$\|\nabla f(X_t)\| \geq 8t_f(\delta)\eta\sigma L^2 \frac{1}{r} \sum_{i=1}^r \|P_{t,i}\|^3.$$

Denote

$$\mathcal{E}_T(\delta) := \mathcal{H}_T(\delta) \cap \left(\bigcap_{t=0}^{T-1} \mathcal{A}_t(\delta) \right) \cap \left(\bigcap_{t=0}^{T-1} \mathcal{G}_t(\delta) \right) \cap \left(\bigcap_{k=0}^{K-1} \mathcal{B}_{kt_f(\delta)}(\delta; t_f(\delta)) \right)$$

For the rest of the proof assume we are working on $\mathcal{E}_T(\delta)$. By Lemma ??, and our choices of η and δ in the statement of the theorem, we have

$$\begin{aligned}f(X_T) - f(X_0) &\leq - \sum_{k \in I_1} \frac{\eta}{4} \min_{t \in J_k} \|\nabla f(X_t)\|^2 + T\eta^4 \frac{cdm\chi^4}{r} t_f(\delta)^2 \sigma^2 L^4 (dm)^3 \log(T/\delta)^6 \\ &\quad + T C c^4 \eta \sigma^2 L^2 (dm)^2 \log\left(\frac{T}{\delta}\right)^4 + \eta^2 T C c^6 \sigma^2 L^3 (dm)^3 \log\left(\frac{T}{\delta}\right)^6\end{aligned}$$

Assume that there are more than $T/4$ iterations with $\|\nabla f(X_t)\| \geq \varepsilon$. Let I_ε denote the set of indices k where there is a $t \in J_k$ with $\|\nabla f(X_t)\| \geq \varepsilon$. Thus by pigeonhole principle, $|I_\varepsilon| \geq T/4t_f(\delta)$. By our choices of parameters, it can be shown that

$$16t_f(\delta)\eta\sigma L^2 c(dm)^{3/2} \log(T/\delta)^3 < \varepsilon$$

While by Lemma ??, if $k \in I_1^c$,

$$\|\nabla f(X_t)\| \leq 16t_f(\delta)\eta\sigma L^2 c(dm)^{3/2} \log(T/\delta)^3$$

Therefore, $I_\varepsilon \subset I_1$. Now by Lemma ??, for any $k \in I_1$,

$$\frac{1}{2} \left\| \nabla f(X_{kt_f(\delta)}) \right\| \leq \left\| \nabla f(X_t) \right\| \leq 2 \left\| \nabla f(X_{kt_f(\delta)}) \right\|, \quad \forall t \in J_k$$

This implies that for any $k \in I_\varepsilon$, $\min_{t \in J_k} \|\nabla f(X_t)\|^2 \geq \frac{1}{16}\varepsilon^2$, and consequently,

$$-\sum_{k \in I_1} \frac{\eta}{4} \min_{t \in J_k} \|\nabla f(X_t)\|^2 \leq -\sum_{k \in I_\varepsilon} \frac{\eta}{2} \cdot \frac{\varepsilon^2}{16} \leq -\frac{T}{4t_f(\delta)} \cdot \frac{\eta}{4} \cdot \frac{\varepsilon^2}{16} = -\frac{T\eta\varepsilon^2}{256t_f(\delta)}$$

Hence,

$$\begin{aligned} f(X_T) - f(X_0) &\leq -\frac{T\eta\varepsilon^2}{256t_f(\delta)} + T\eta^4 \frac{cdm\chi^4}{r} t_f(\delta)^2 \sigma^2 L^4(dm)^3 \log(T/\delta)^6 \\ &\quad + TCc^4\eta\sigma^2 L^2(dm)^2 \log\left(\frac{T}{\delta}\right)^4 + \eta^2 TCc^6\sigma^2 L^3(dm)^3 \log\left(\frac{T}{\delta}\right)^6 \end{aligned}$$

By our choice of parameters,

$$\begin{aligned} &T\eta^4 \frac{cdm\chi^4}{r} t_f(\delta)^2 \sigma^2 L^4(dm)^3 \log(T/\delta)^6 + TCc^4\eta\sigma^2 L^2(dm)^2 \log\left(\frac{T}{\delta}\right)^4 \\ &+ \eta^2 TCc^6\sigma^2 L^3(dm)^3 \log\left(\frac{T}{\delta}\right)^6 \leq \frac{T\eta\varepsilon^2}{512t_f(\delta)} \end{aligned}$$

We thus get

$$f(X_T) - f(X_0) \leq -\frac{T\eta\varepsilon^2}{512t_f(\delta)}$$

Therefore, as long as

$$T \geq \frac{512t_f(\delta)(f(x_0) - f^*)}{\eta\varepsilon^2}$$

we would get $f(X_T) < f^*$, a contradiction. Thus, we may conclude that on the event $\mathcal{E}_T(\delta)$ there are at most $T/4$ iterations for which $\|\nabla f(X_t)\| \geq \varepsilon$.

Finally,

$$\begin{aligned}\mathbb{P}(\mathcal{E}_T^c) &\leq \mathbb{P}(\mathcal{H}_T^c) + \sum_{t=0}^{T-1} \mathbb{P}(\mathcal{A}_t(\delta)) + \sum_{t=0}^{T-1} \mathbb{P}(\mathcal{G}_t(\delta)) + \sum_{k=0}^{K-1} \mathbb{P}\left(\mathcal{B}_{kt_f(\delta)}(\delta; t_f(\delta))\right) \\ &\leq \frac{(T+2)\delta}{T} + \delta + \delta + \frac{K\delta}{T} \leq 5\delta.\end{aligned}$$

□

Chapter 6

Conclusion

6.1 Summary

Zeroth-order optimization is prohibitively expensive in high dimensions, typically needing d times more function evaluations than gradient descent. If we parallelize function evaluations across r workers, we can typically reduce the iterations by a factor of r , but before this work, the theory assumed perturbations sampled were Gaussian in every entry. EGGROLL replaces these full Gaussian perturbations with structured low-rank matrices $ab^\top$ for Gaussian vectors a, b (?). Suddenly, zeroth-order algorithms were practical. But it remained an open question whether these algorithms actually converge.

EGGROLL Convergence. In Chapter ??, we answered the above question in the affirmative, proving that, after a careful choice of parameters, EGGROLL really does converge to a first-order stationary point for any L -smooth (possibly nonconvex) function f . The principal theoretical contribution of that chapter was Theorem ??, showing the typical size of the gradient when using low-rank perturbations $ab^\top$, is the same as for Gaussian perturbations, up to some irrelevant log factors, vastly generalizing Proposition 2 of ?.

Having established this lemma, we can prove that EGGROLL converges.

Zeroth-Order Mirror Descent. Perturbed Gradient Descent was shown to rapidly escaping saddle points with only a log penalty on dimension (?). ? showed that perturbed Zeroth-Order Gradient Descent can escape saddle points too, in time $O(1/\varepsilon^{5/2})$. Having established EGGROLL converges, rather than just generalize perturbed gradient descent to the low-rank case, we sought a rate better than $O(1/\varepsilon^{5/2})$, preferably the sub-quadratic $O(1/\varepsilon^{7/4})$ found in ? and ?. ? design a very general method which relies on two subroutines: accelerated gradient descent and finding the top Hessian eigenvector. In pursuit of a zeroth-order gradient descent, we first showed that zeroth-order mirror descent converges in Chapter ?? with the typical d zeroth-order penalty for quadratic convex functions.

Zeroth-Order Accelerated Gradient Descent. Having established the theoretical tools needed for proving convergence of zeroth-order mirror descent, we use the linear mixing framework introduced by ? to design a zeroth-order accelerated gradient descent algorithm, which converges with the typical d zeroth-order penalty, with high probability, in a single run. It is a high-probability analysis and not an expected one, no independent trials are needed. We precisely quantify how an inexact gradient oracle propagates error through the algorithm, by bounding the martingale term by other $\Theta(1)$ terms already present. We believe the techniques developed in this Chapter to be generalizable to other problems which need to bound a martingale difference sequence.

Impact. Armed with the ability to bound mean-0 martingale error terms and the typical scale of the EGGROLL gradient estimator with high probability, many high-probability proofs in zeroth-order descent can be generalized to this low-rank perturbation setting.

Low-rank directions are especially suitable for matrix-valued neural network parameters, substantially reducing memory requirements from $O(rdm)$ to $O(r(d+m))$, allowing for the use of zeroth-order algorithms in practice (?). By parallelizing function evaluations across r workers, we converge to a first-order stationary point r times faster (Chapters ??, ??, ?). Thus, a twofold increase in compute could lead to a two-times decrease in training time. As investment in compute continues to grow, these results suggest zeroth-order methods could become a useful complement to gradient-based reinforcement-learning methods such as PPO and GRPO.

6.2 Limitations

Optimizing σ . Here, we have not attempted to optimize the smoothing radius σ . This is important as a larger σ typically correlates with a more stable algorithm. Furthermore, when function evaluations are stochastic with fixed additive noise μ , one has to balance the error from σ with a μ/σ introduced by the gradient estimator. So for stochastic functions, larger σ is better.

Logarithms. We have ignored logarithms throughout this work. It might be possible to reduce logarithmic dependence on ε in Chapter ??, but we have not attempted it here. While common in analysis of randomized algorithms to ignore logarithmic dependence on problem parameters, some work has focused on removing this (?).

Function-Evaluation Complexity. While we have shown that parallelizing function evaluations to r workers reduces wall-clock time by a factor of r , the total number of function evaluations is still d times larger than gradient descent. No work in the literature has gotten a sublinear dependence on d for function evaluations for optimizing convex

functions in the zeroth-order setting.

Logarithmic Lower Bound on r . In Chapter ??, we assume a lower bound $r \geq \tilde{O}(1)$. Significant work in ? and Chapter ?? went into removing this assumption. However, we expect it to be necessary for this zeroth-order linear coupling framework, due to the nature of restarting z_0 back to its original value every restart. In effect, z_0 and x_0 could drift permanently far apart, preventing us from establishing that gradients change by at most a factor of 4 over an interval of log-length, unlike Lemma ??.

L -Smoothness Assumption All our results depend on an L -smoothness assumption. This is more general a Lipschitz assumption, which frequently does not hold for ML models, but it still requires gradients to not change too quickly in space. In the EGGROLL paper ?, they assume only a local α -Holder condition on gradients, which is more relaxed than L -smoothness.

6.3 Future Work

Experiments. This is a purely theoretical work, no experiments were performed. In a future work, significant experimental work should go into tuning α and η for zeroth-order linear mixing and zeroth-order convergence to a FOSP (Chapters ?? and ??). For large ML models, we typically only use batched-loss, which makes function evaluations stochastic, not deterministic like we have assumed here. This might necessitate a variance-reduction technique, or adaptive choices for η and α .

Anisotropic Perturbations. We have assumed throughout that perturbations are sampled with covariance I . However, there has been some work on using anisotropic perturbations, see ?. The best anisotropic estimator would have covariance $\nabla f(x)\nabla f(x)^\top$

at x , since we could immediately recover the gradient direction. Perhaps remembering a time-average of past gradients ($\tau < t$) could inform the gradient at time t , thereby reducing the variance of the estimator. There has been some preliminary work in this direction (?, ?).

Generalizing Existing ZO methods. Armed with Theorem ?? and its consequences, we suspect many existing ZO methods can be generalized to the case of low-rank perturbations. We have pursued this for ?, but we suspect there is more to do here.

6.4 Concluding Remarks

Low-rank perturbations are a powerful tool for making zeroth-order algorithms practical for neural networks. This work shows this low-rank structure still permits rigorous high-probability guarantees, rather than being merely an engineering device. We’ve shown how zeroth-order accelerated convex and non-convex optimization can be analyzed with high probability, quantifying propagated error through a mean-zero martingale error terms. We’ve shown how the population size r trades parallel compute for a decrease in wall-clock time. While other fine-tuning methods like GRPO and PPO are only partially understood theoretically, we establish that EGGROLL converges to a FOSP in the deterministic L -smooth setting. Low-rank perturbations can be used throughout zeroth-order optimization, possibly allowing for the practical deployment of these algorithms. This dissertation provides a theoretical foundation for these low-rank perturbations and a starting point for future work.

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